



**FAPS**

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# **Analysis of the process chain for manufacturing of electrical drives**

Advanced Lecture Unit 10

**This training unit provides a repeated overview of the main technologies and methods in the production process chain of electrical machines.**

**The production of electric drives involves a multitude of usable process possibilities. It is therefore essential to learn a holistic overview of the individual processes and their appropriate linking in the form of process chains.**

**After attending the training unit "Analyzing the process chain for manufacturing of electric drives", you should:**

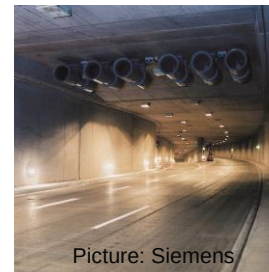
- have repeated and internalized the processes and process chains learned in the lecture
- Be able to set up and evaluate process chains using specific examples



Picture: Omron



Picture: ZF Friedrichshafen



Picture: Siemens



Picture: Kuka

The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

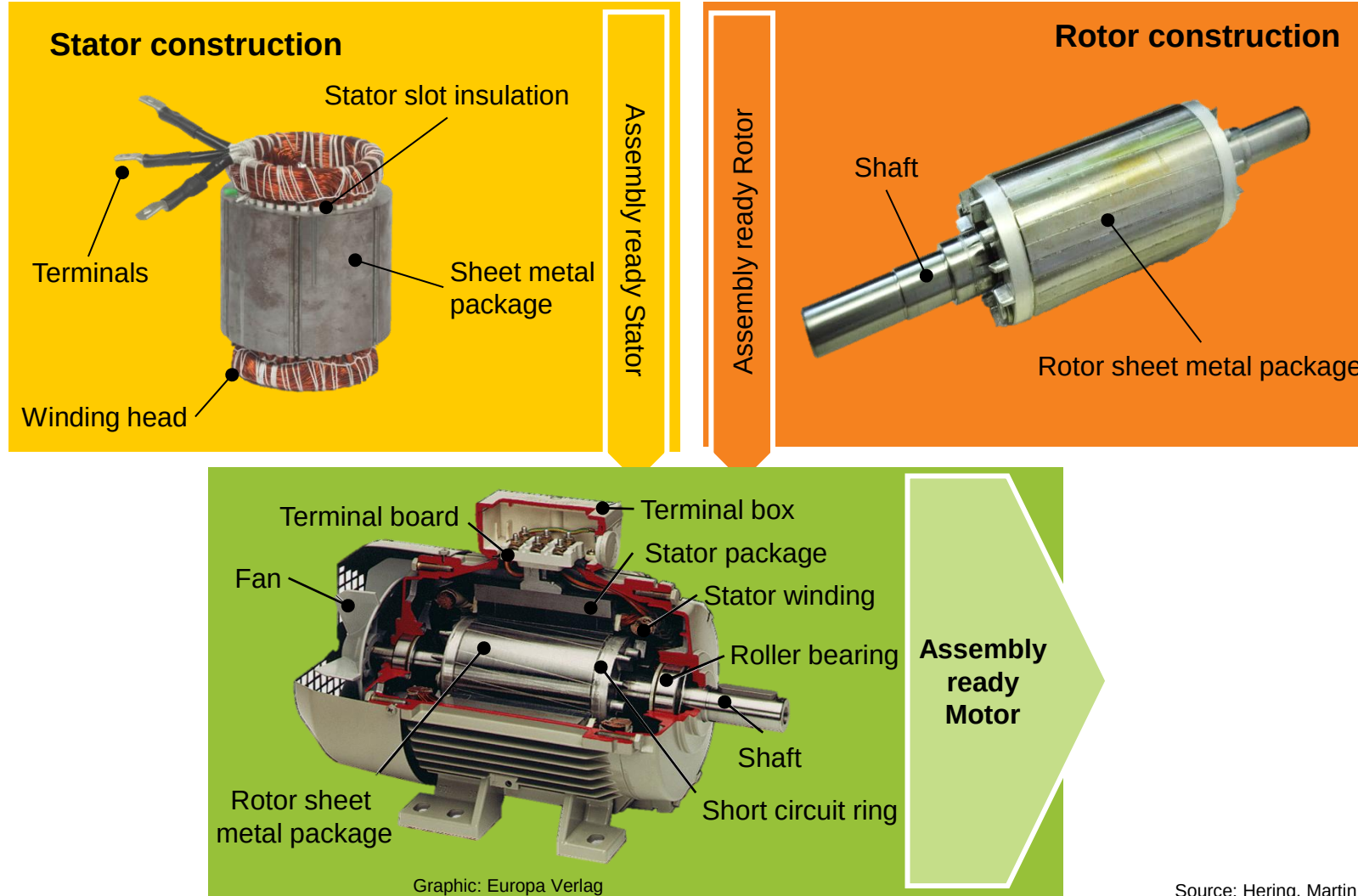
- **Repetition of the production process chains**
  - Higher-level production process chain of an electric motor
  - Production of the soft magnetic core
  - Insulation of the slot base
  - Winding process
  - Electrical contacting
  - Insulation systems
  - Permanently excited rotors
  - Mechanical components of electric motors
- Process chain analysis based on practical examples



Picture: VW

Rotating electric machines consist of a stationary part - the stator, and a rotating part - the rotor.

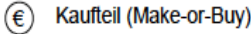
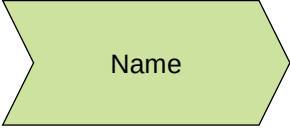
Slide from LU1



- The stator serves to create the rotating magnetic field.
- The rotor converts a magnetic moment into a mechanical torque and transfers power to the shaft.
- The housing serves to supply and protect the motor.

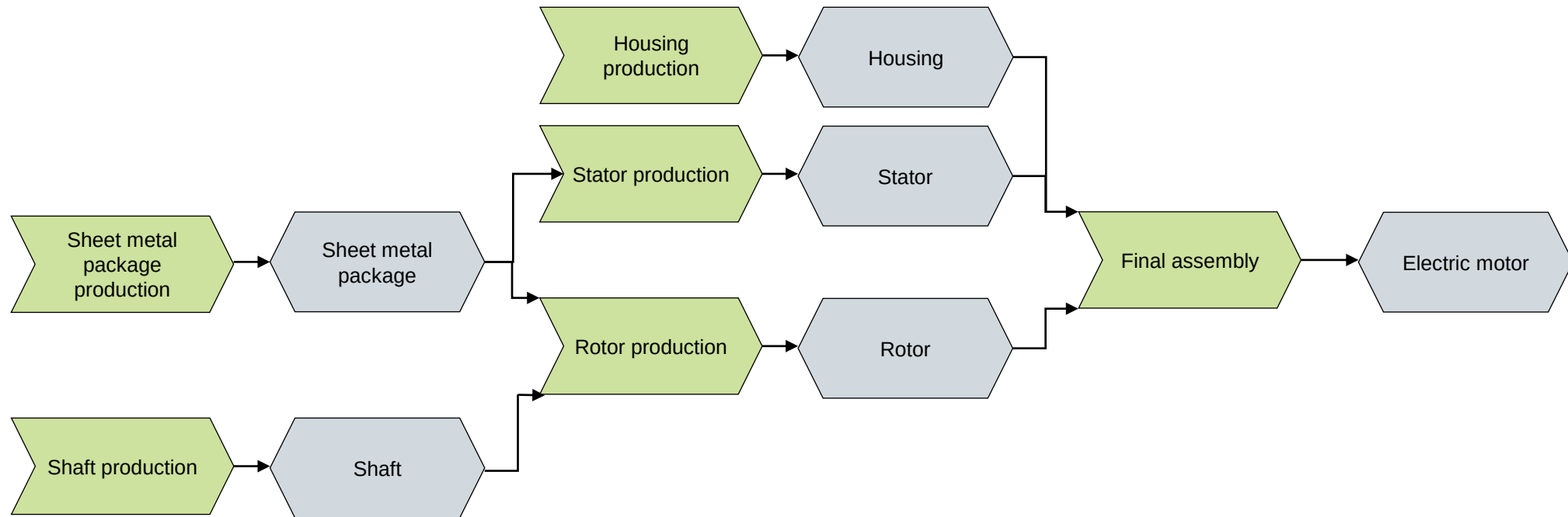
Source: Hering, Martin et al. 2012 – Elektrotechnik und Elektronik für Maschinenbauer

The process chain is illustrated in the lecture based on the notation elements of CONSENS.

|                                       | Elements                                                                                                      | Relationships                                                                                          | Pictograms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Extension                                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Display elements of the process chain | <b>Material elements</b><br> | <b>Sequence</b><br> | <b>Material element differentiation</b><br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Logical operators:</b><br><br>Exclusive Or  |
|                                       | <b>Processes</b><br>         |                                                                                                        | <b>Element-related information</b><br> Parameter  Shape<br><b>System-related information</b><br> Reference to shape<br> Reference to requirement<br> Reference to process<br> Reference to resource<br> Reference to material | Inclusive Or                                   |

The process chain for manufacturing an electric motor comprises a large number of process steps and material elements. A manufacturing process is followed by a material element.

Slide from LU1

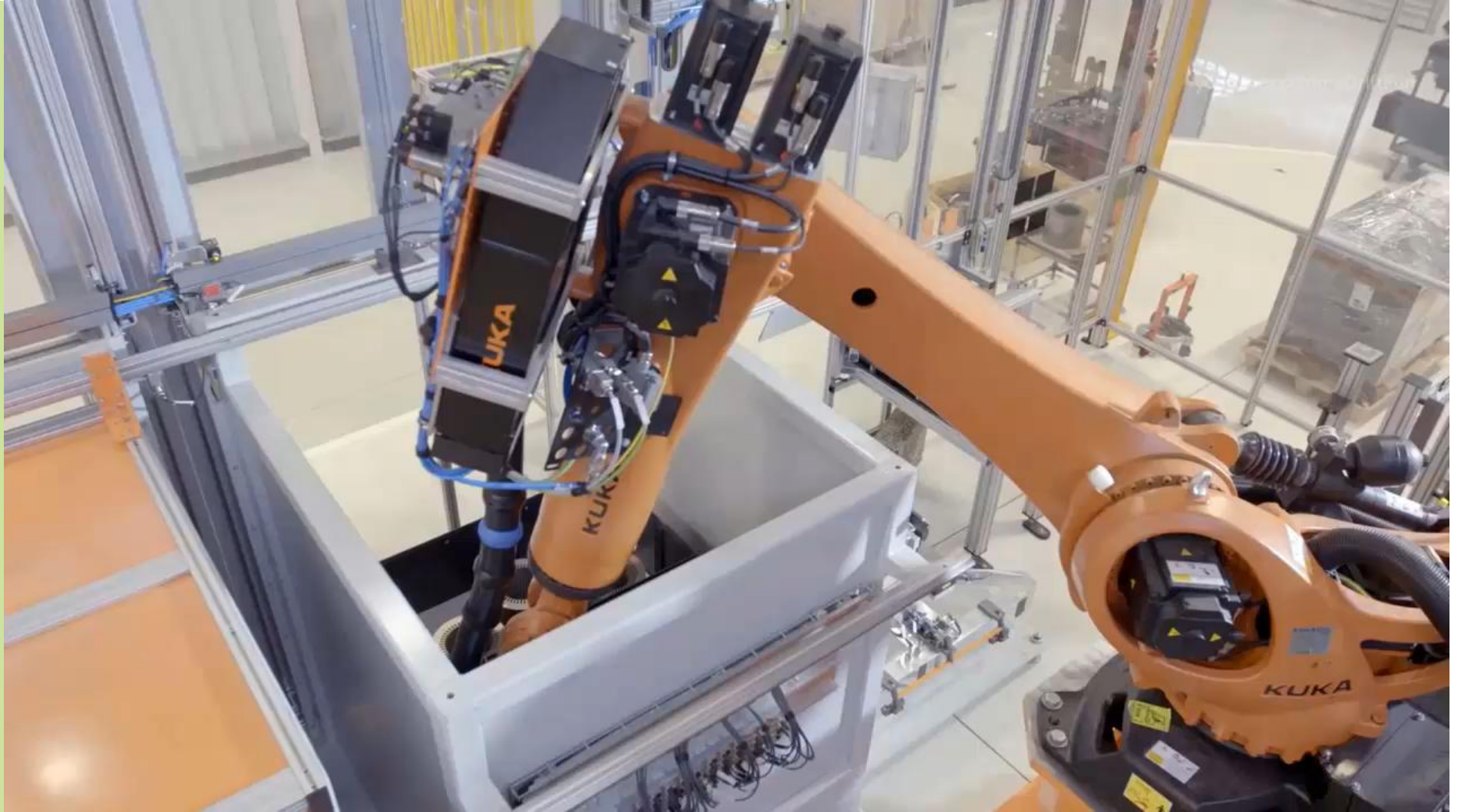


The automated production of an electric drive is widely used in the automobile industry.

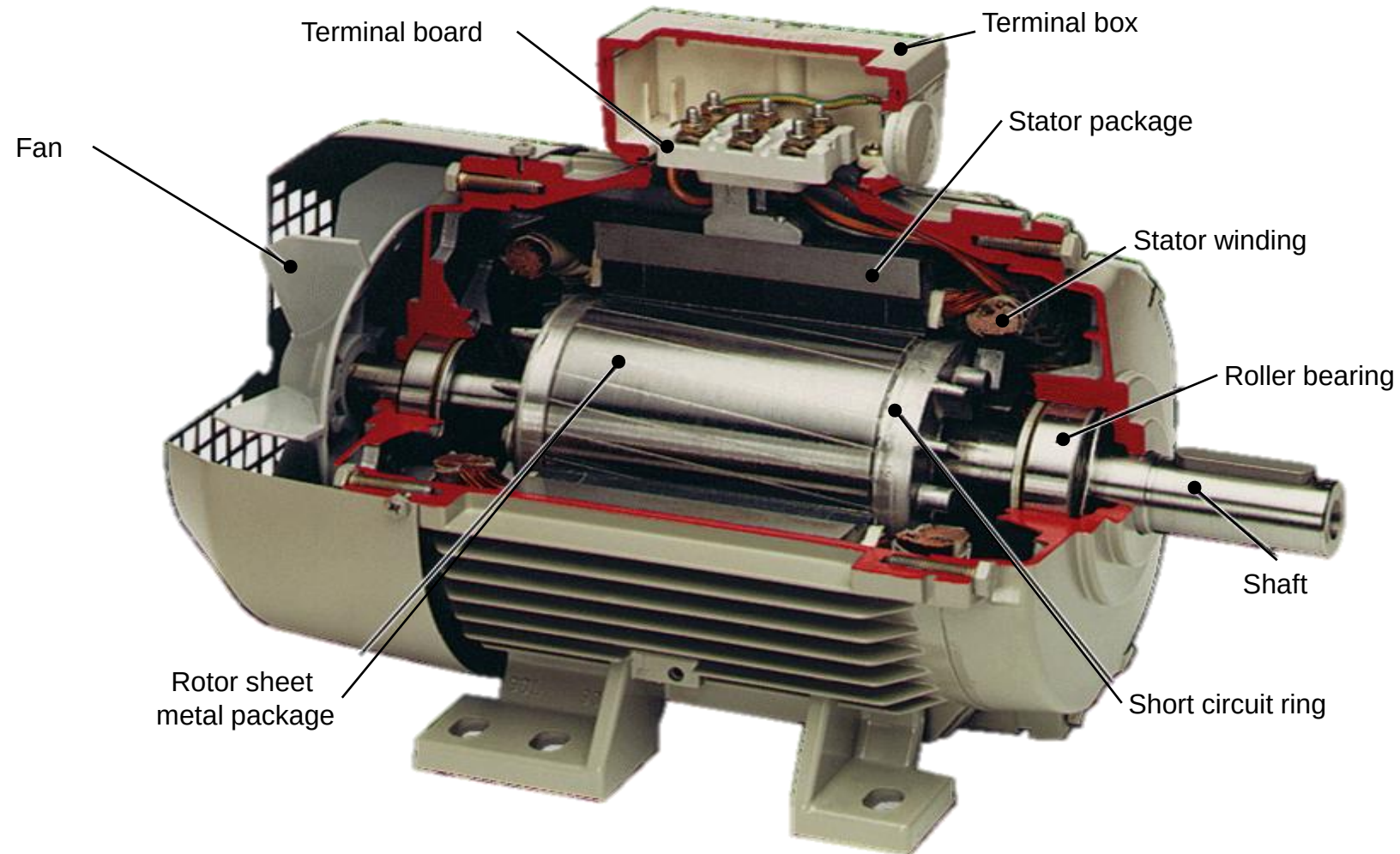
## Process

Automated assembly of the windings  
of an electric drive

Video: Audi AG

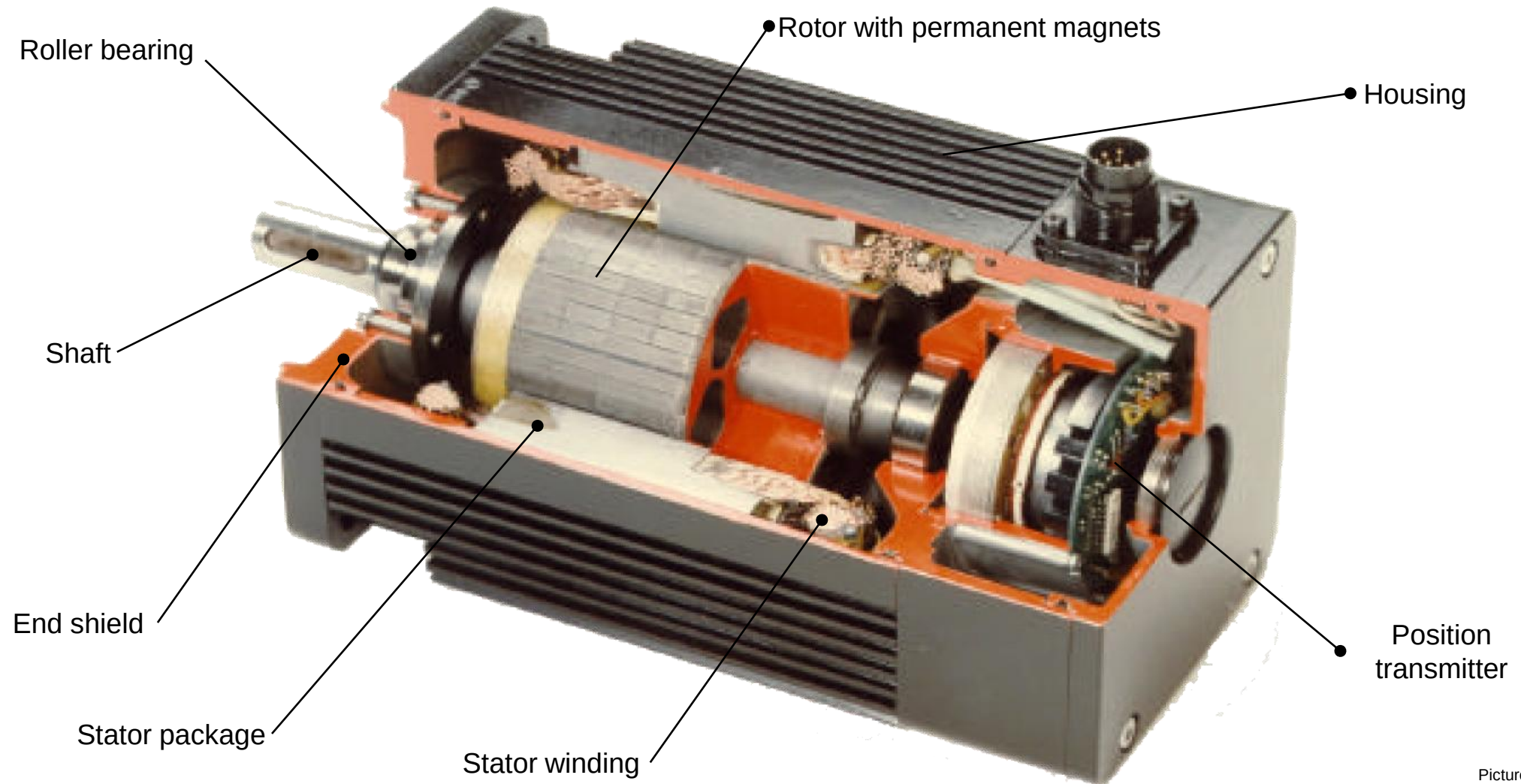


In asynchronous motors, the excitation voltage is induced in the short-circuit windings of the rotor.



Picture: Europa Verlag

**Synchronous motors always require upstream electronics.**



Picture: Bosch

The "carbon brushes" of a DC machine slide over the commutator, so that they have to be replaced due to wear.

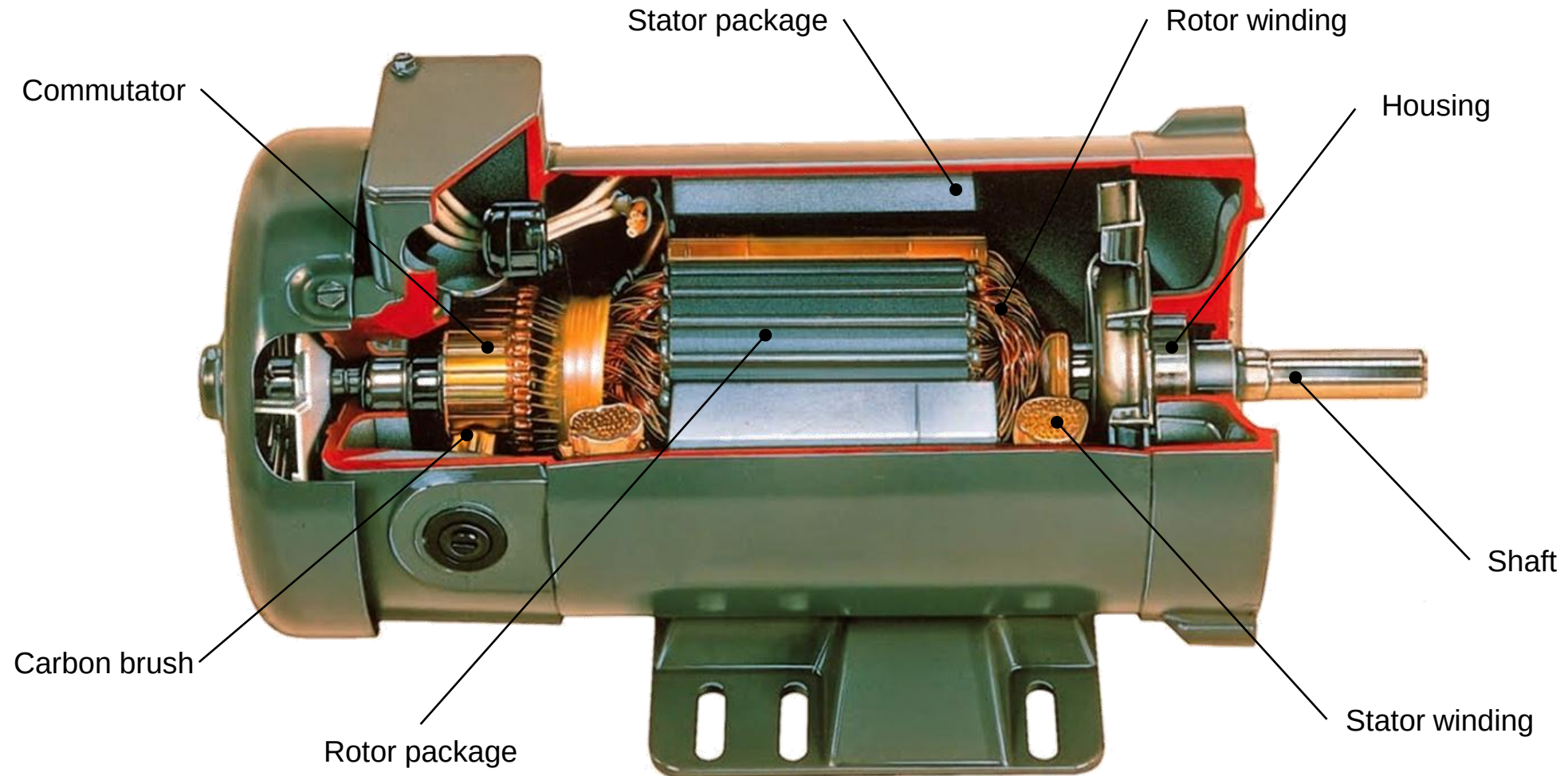


Image: Courtesy Reliance Electric Co.

The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

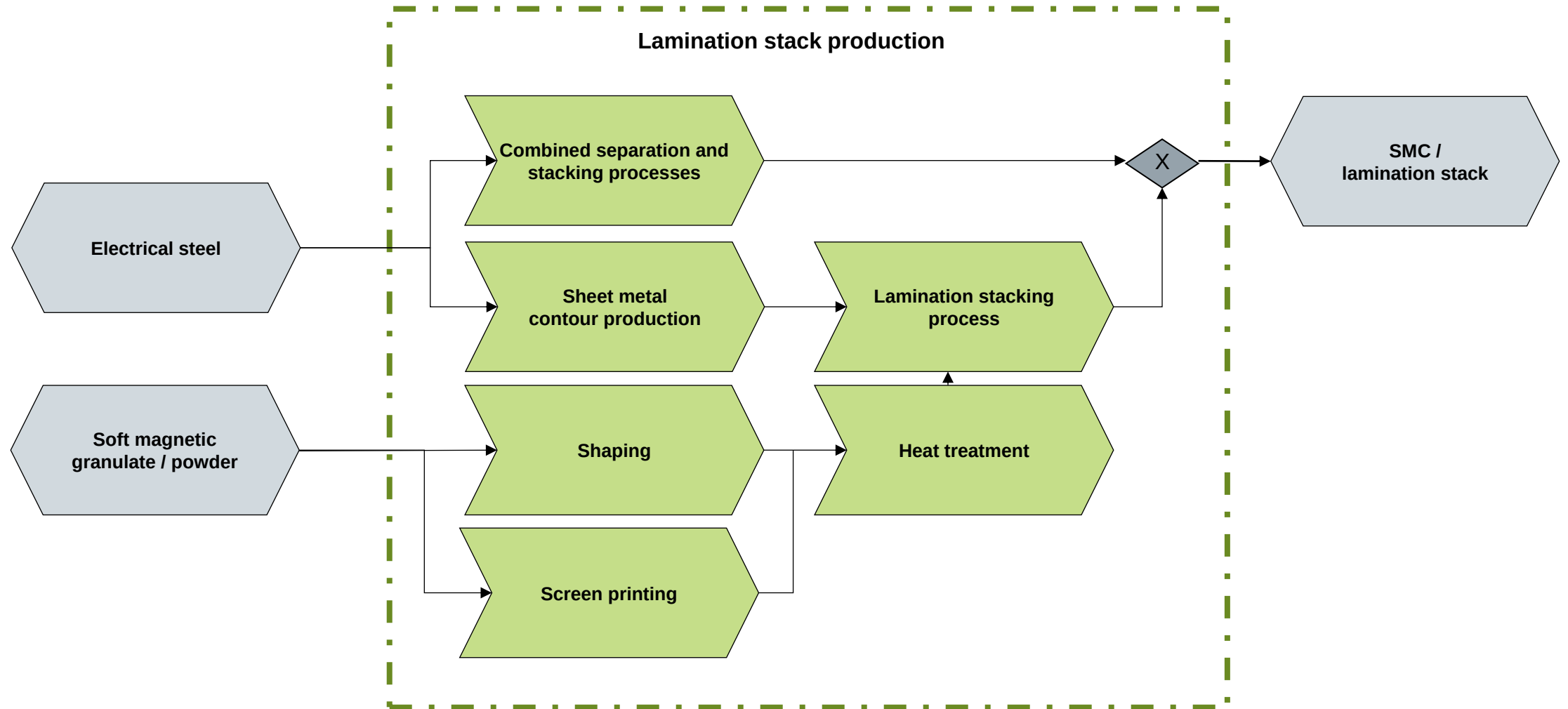
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Picture: VW

The soft magnetic core consists either of layered electrical steel sheets or from sintered powder composite materials.

Slide from LU3



Large presses with expensive tools are necessary for combined stamping and packaging.

### Process

Automated production of sheet metal packages using the punching stacking process

Video: Ing Yu



Laser cutting of sheet metal is preferably used in prototype production.

## Process

Automated laser cutting of electrical sheets for prototype production

Video: FAPS



## Welded sheet metal stacks fulfill high mechanical requirements.

### Process

Automated arc welding of electrical sheets

Video:  
FASP

[youtube.com/watch?v=mQrS-\\_R6GvQ](https://youtube.com/watch?v=mQrS-_R6GvQ)



The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

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Picture: VW

The insertion of the slot base insulations into the stator is usually fully automatic.



## The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

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Picture: VW

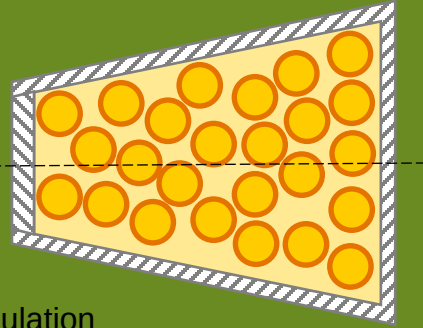
When talking about fill factors, it is important to pay close attention to the definition used due to the different calculation methods.

### Slot fill factor

$$k_N = \frac{z_n \cdot d_i^2}{A_{ni}}$$

with

- $z_n$  – Number of wires per slot
- $d_i$  – Outer diameter of wire and insulation
- $A_{ni}$  – Cross section of the insulated slot

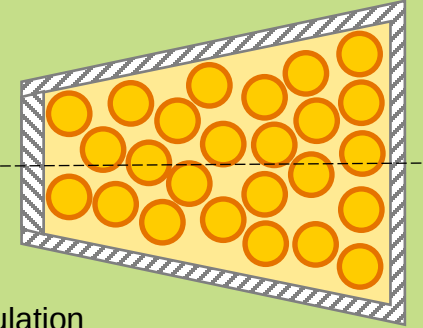


### Mechanical fill factor

$$k_{mech} = \frac{A_{Wire,i}}{A_{ni}} = \frac{\frac{1}{4} \cdot d_i^2 \cdot \pi \cdot z_n}{A_{ni}}$$

with

- $z_n$  – Number of wires per slot
- $d_i$  – Outer diameter of wire and insulation
- $A_{ni}$  – Cross section of the insulated slot



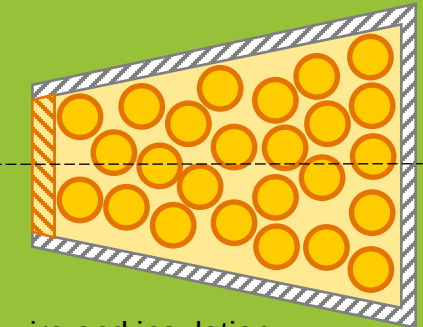
## Fill factor

### Winding fill factor

$$k_W = \frac{z_n \cdot d_{i\max}^2}{A_{neff}}$$

with

- $z_n$  – Number of wires per slot
- $d_{i\max}$  – Largest outer diameter of wire and insulation
- $A_{neff}$  – Cross section of the insulated slot and slot opening

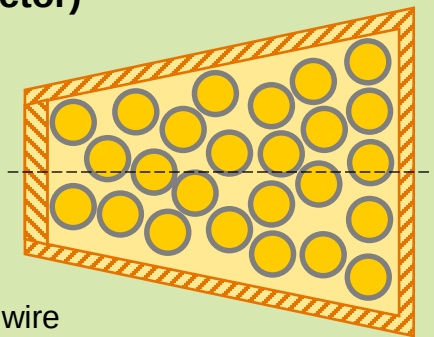


### Electrical fill factor (copper fill factor)

$$k_{elec} = \frac{\frac{1}{4} \cdot d_{Cu}^2 \cdot \pi \cdot z_n}{A_n}$$

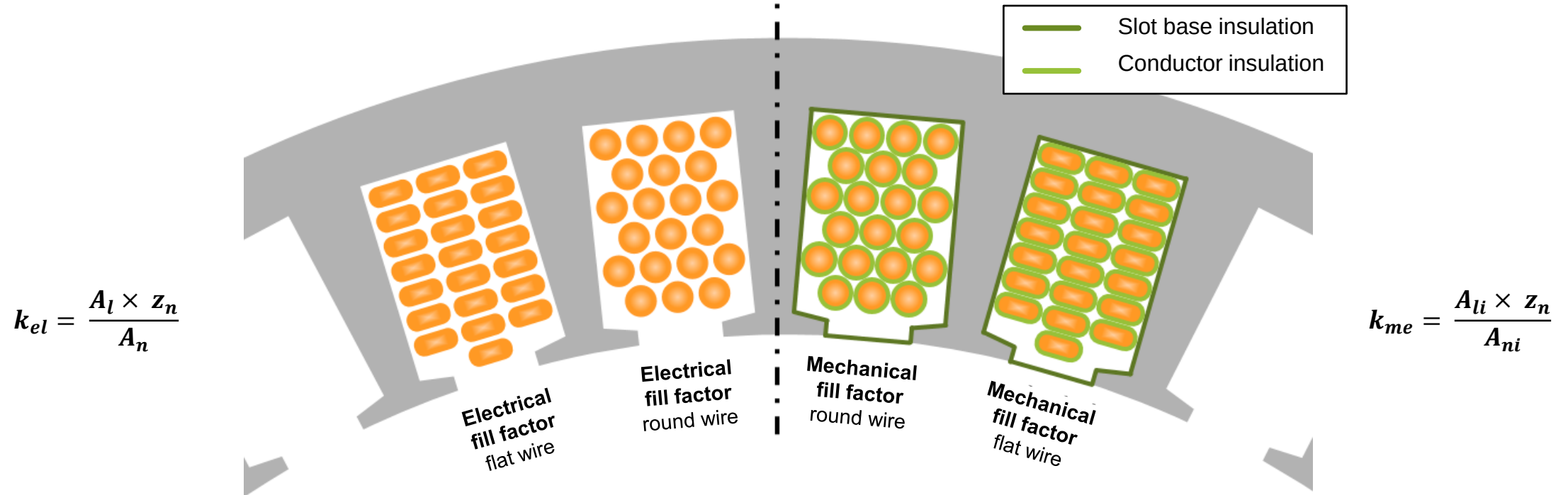
winding

- $z_n$  – Number of wires per slot
- $d_{Cu}$  – Nominal diameter of the bare wire
- $A_n$  – Total slot cross section



The fill factor is a measure of the proportion of conductive material in relation to the to the area of the slot, regardless of the shape of the conductor.

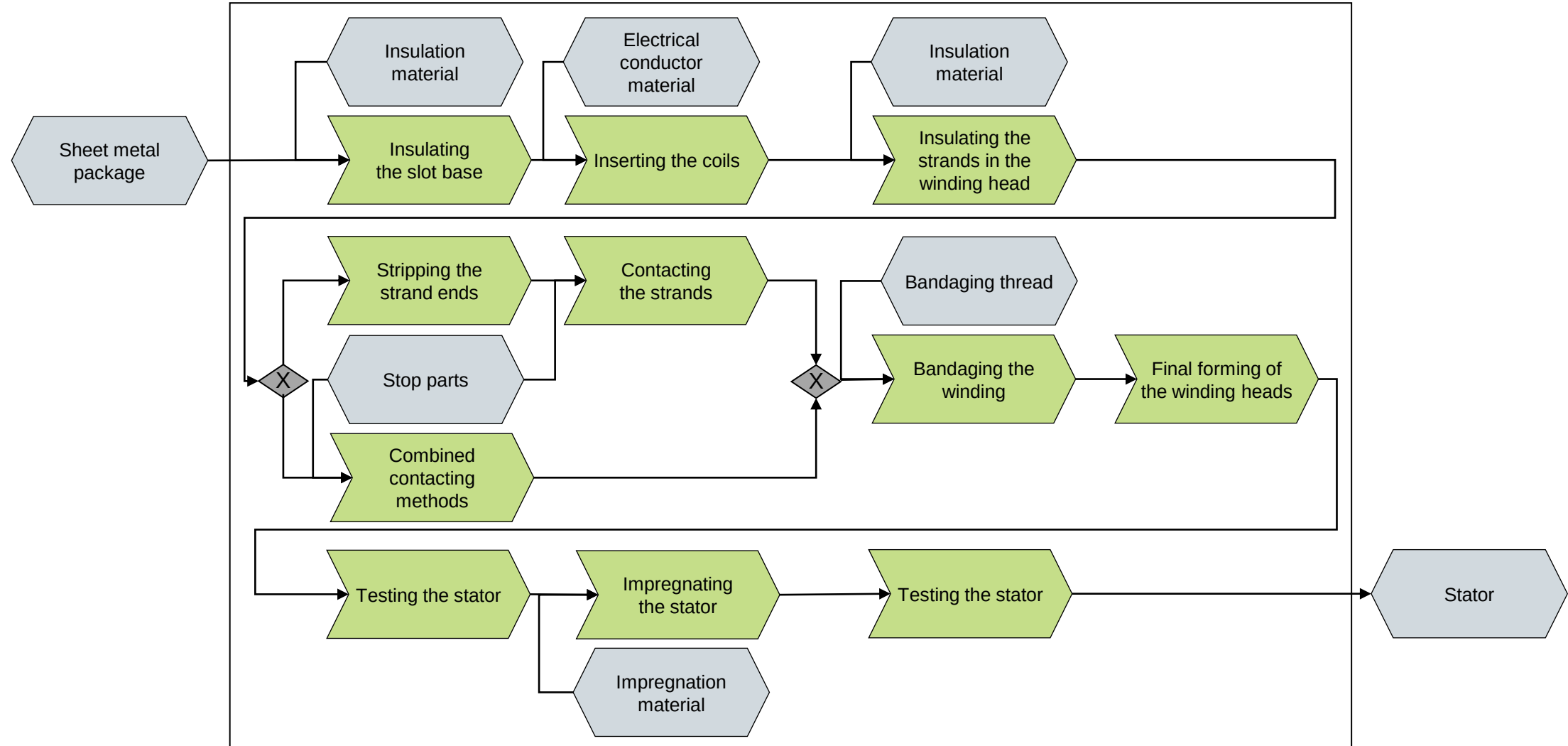
Slide from LU5



|            |                                                          |            |                                                               |
|------------|----------------------------------------------------------|------------|---------------------------------------------------------------|
| $k_{el}$ : | Electrical fill factor (copper fill factor) [%]          | $k_{me}$ : | Mechanical fill factor [%]                                    |
| $A_l$ :    | Cross-sectional area of the bare wire [mm <sup>2</sup> ] | $A_{li}$ : | Cross-sectional area of the insulated wire [mm <sup>2</sup> ] |
| $A_n$ :    | Total slot cross-section [mm <sup>2</sup> ]              | $A_{ni}$ : | Slot cross-section of the insulated slot [mm <sup>2</sup> ]   |
| $z_n$ :    | Number of wires per slot                                 | $z_n$ :    | Number of wires per slot                                      |

Conventional stator production with wound coils comprises a large number of process steps involving the handling of different materials.

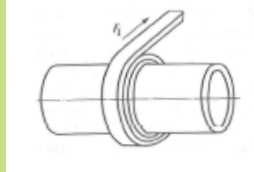
Slide from LU6



In linear winding, the wire is guided onto a rotating coil body by a linear travelling guide element.

### Linear winding process

- Wires fed from a single wire guide
- Coil body rotates about the z-Axis  $\Rightarrow$  radial winding structure
- Linear motion of the wire guide along the z-Axis  $\Rightarrow$  axial winding structure
- Without feed motion, disc coils are produced:
- Optional: Use of guide plates

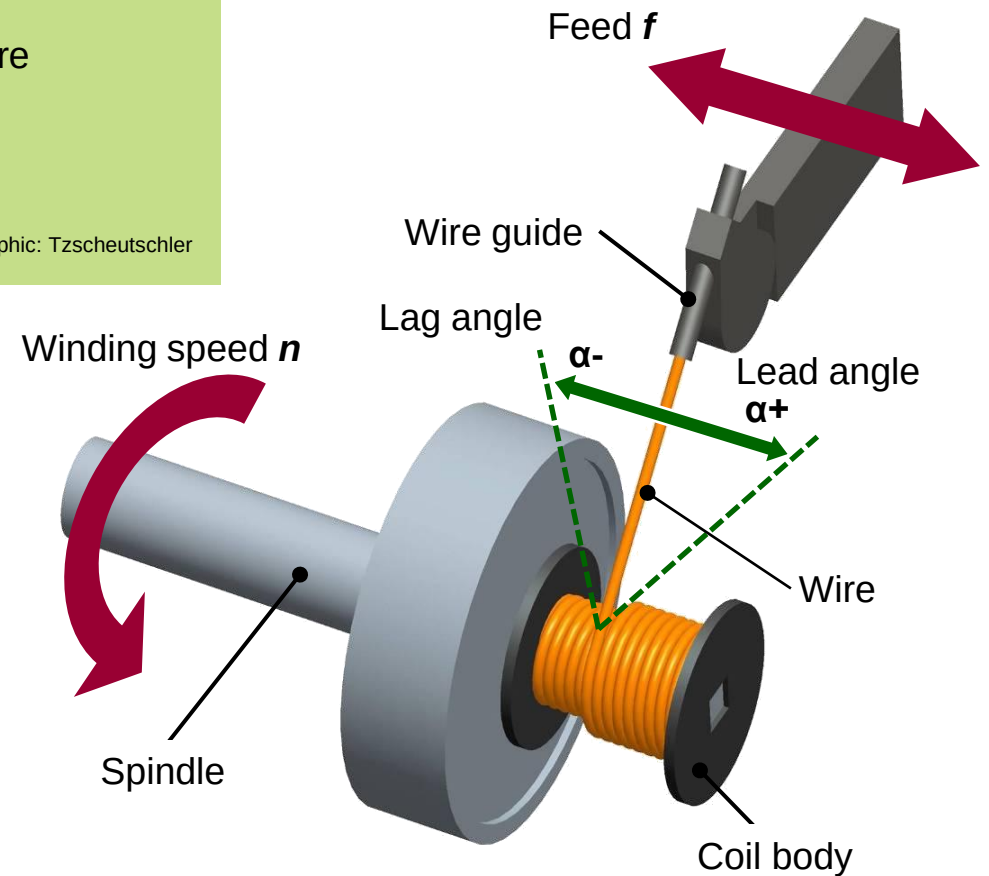


Graphic: Tzscheutschler



Video:  
Aumann

<https://www.youtube.com/watch?v=cq5vRQEQMbW>

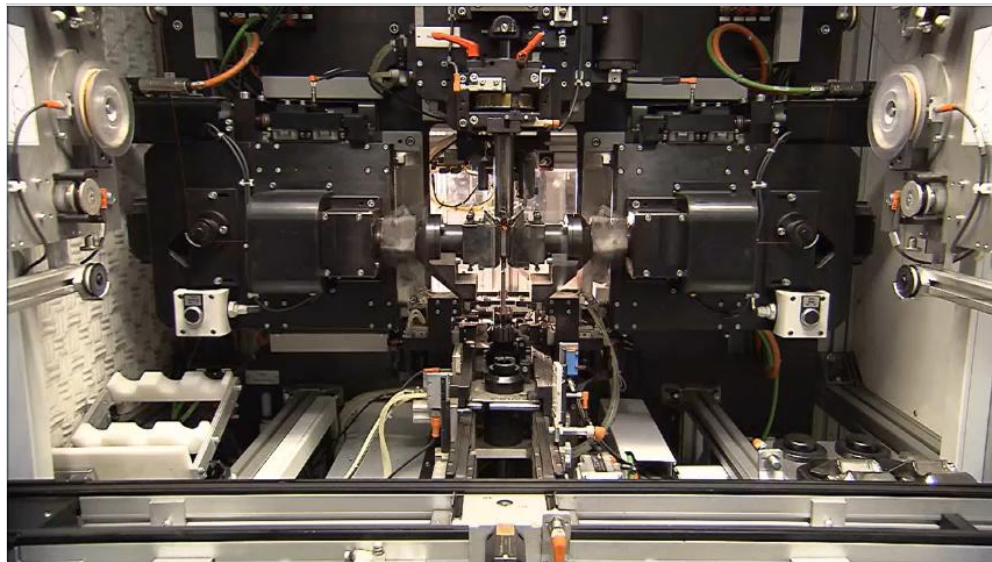


During flyer winding, the wire is guided from a rotating flyer to a fixed winding body.

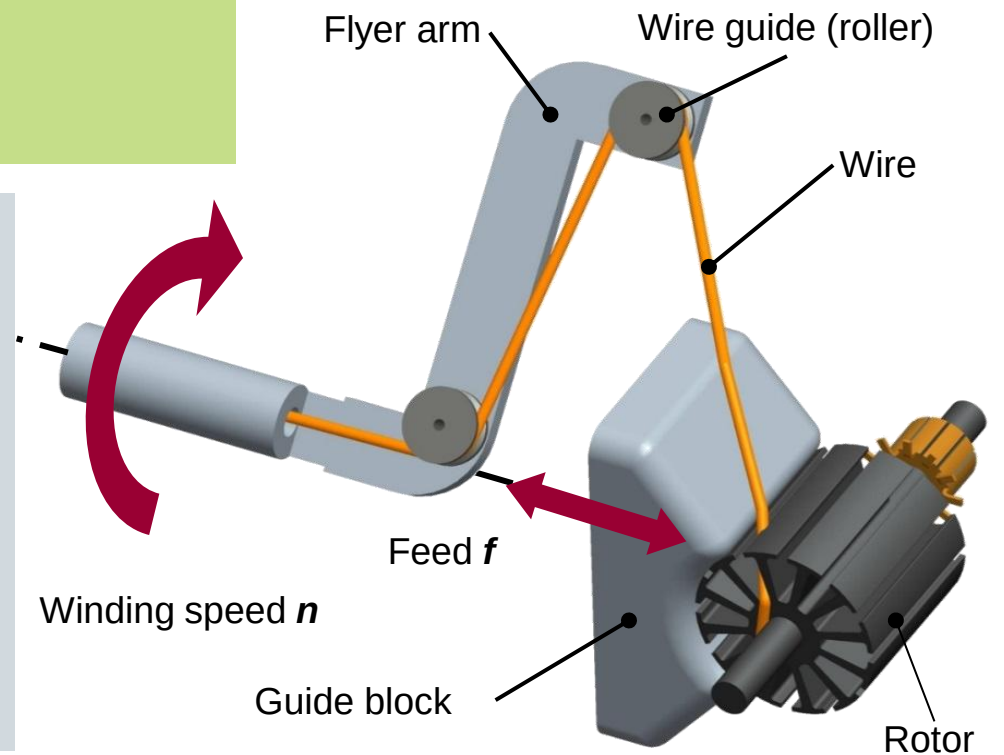
Slide from LU6

### Flyer winding process

- Wire guided over a flyer arm
- Flyer rotates about the z-axis
- Coil body fixed for one coil winding at a time, then rotated by one slot
- Winding wire slides over guide block into the winding chamber
- Optional relative motion in the z-axis



Video:  
Bosch

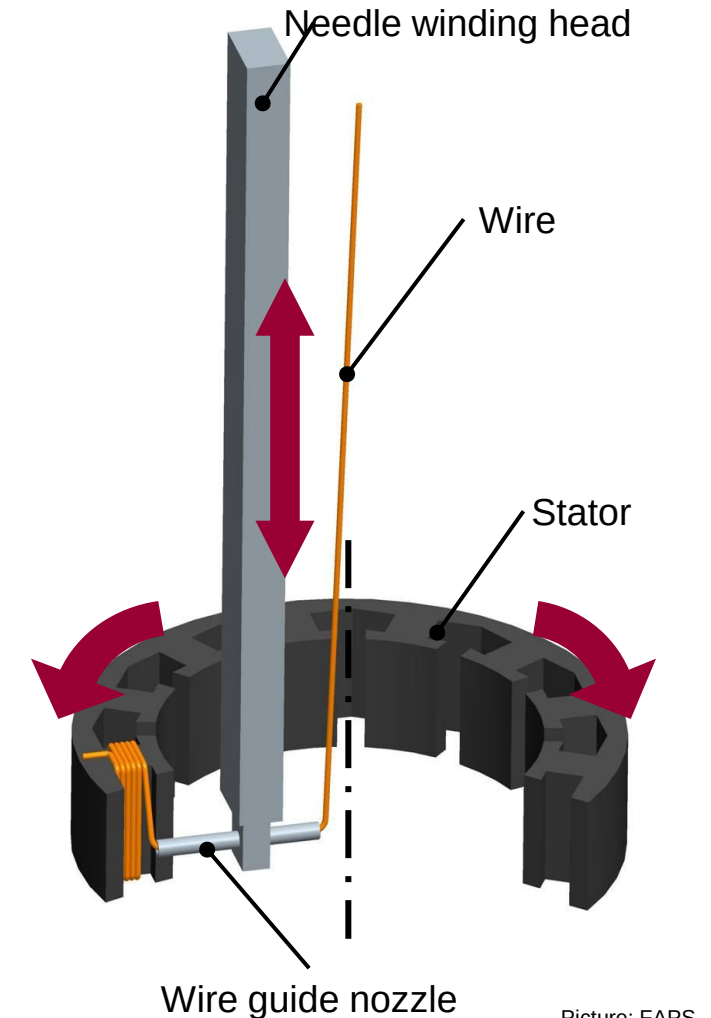
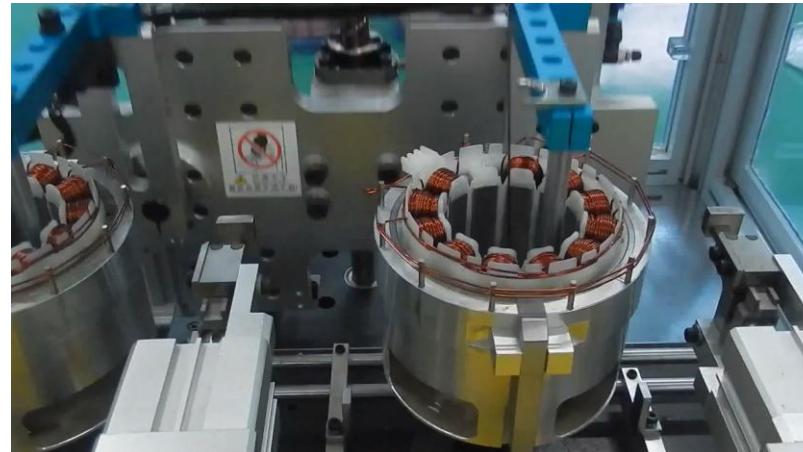


In needle winding, the wire is drawn out using a wire guide nozzle dipping into the slots of the winding body.

Slide from LU6

### Needle winding process

- Wire feed through wire guide nozzle ("needle")
- Stroke movement of the wire guide nozzle through the slot opening  
⇒ Winding formation in the slot
- Pivoting movement of the winding body by one pole pitch  
⇒ Winding formation in the winding head
- Optional pivoting of the needle winding head

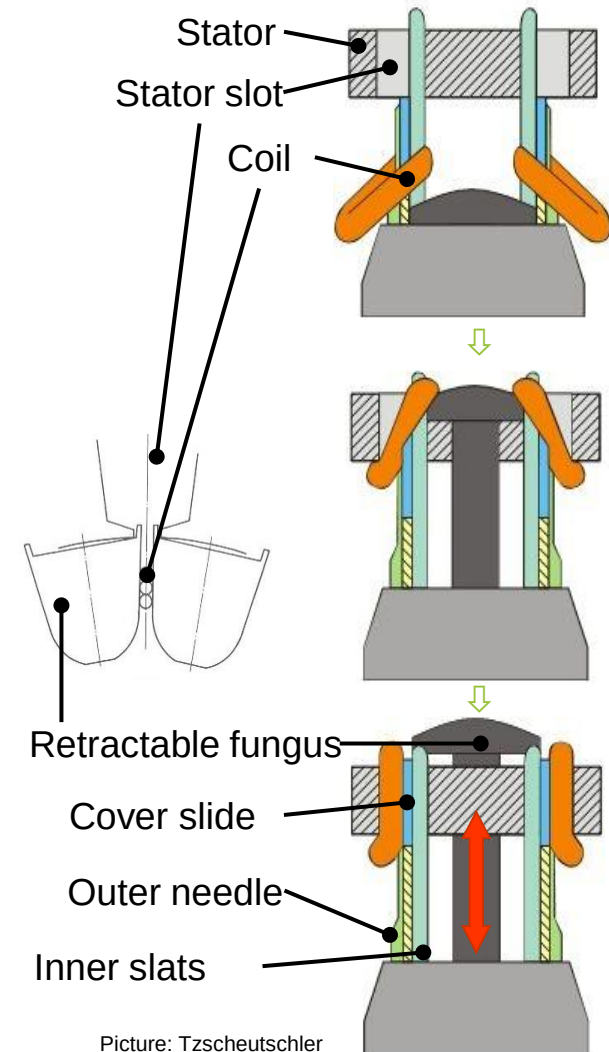


In the Insertion process, the wires of a pre-wound coil are drawn into the slots by means of a insertion piston.

Slide from LU6

### Insertion process

- Coil production in separate process (Linear, Flyer) (or direct on the tool needles)
- Placement of coils between the insertion tool needles
- Placement of the stator onto a setting tool
- Axial stroke of insertion piston exerts pressure onto the coils
- Wires are pulled into the stator slots between the insertion tool inner needles
- Slot closures and phase separators are inserted after the stroke movement



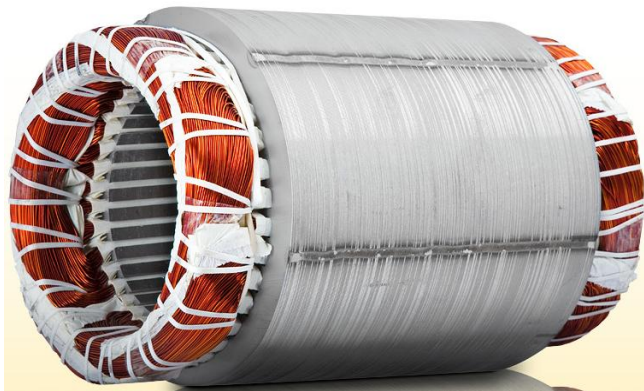
Picture: Tzscheutschler



The shape and especially the height of the winding head are important features of electric drives.

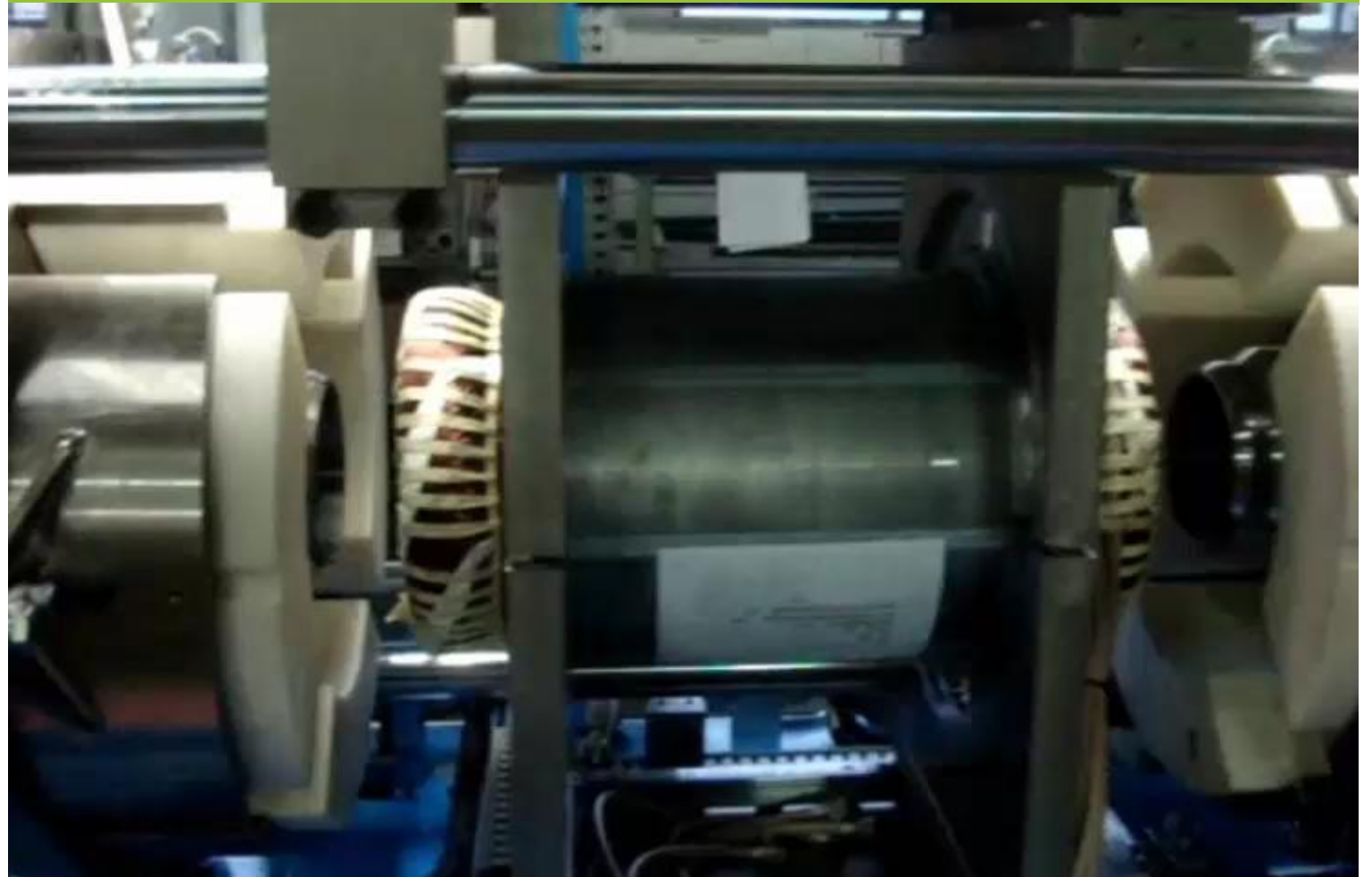
### Pressing the winding head

- Typically used with round wire windings produced with indirect winding processes (insertion process, drip-in process)
- Creates a defined winding head form
- Reduces the space requirement of the winding head

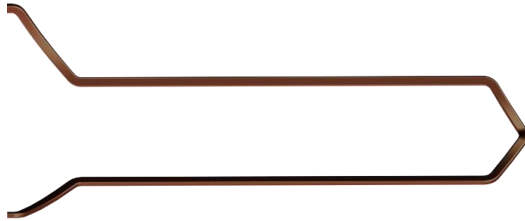
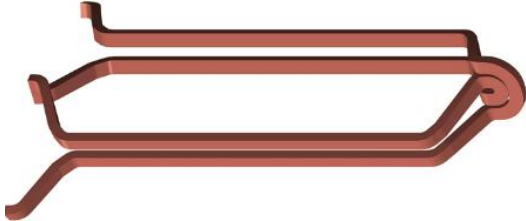
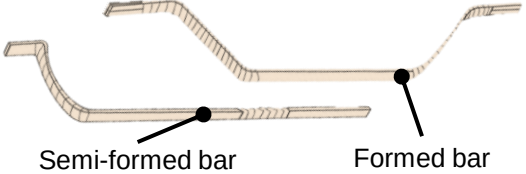
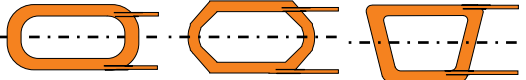


Picture: Renke & Müller GmbH

### Process

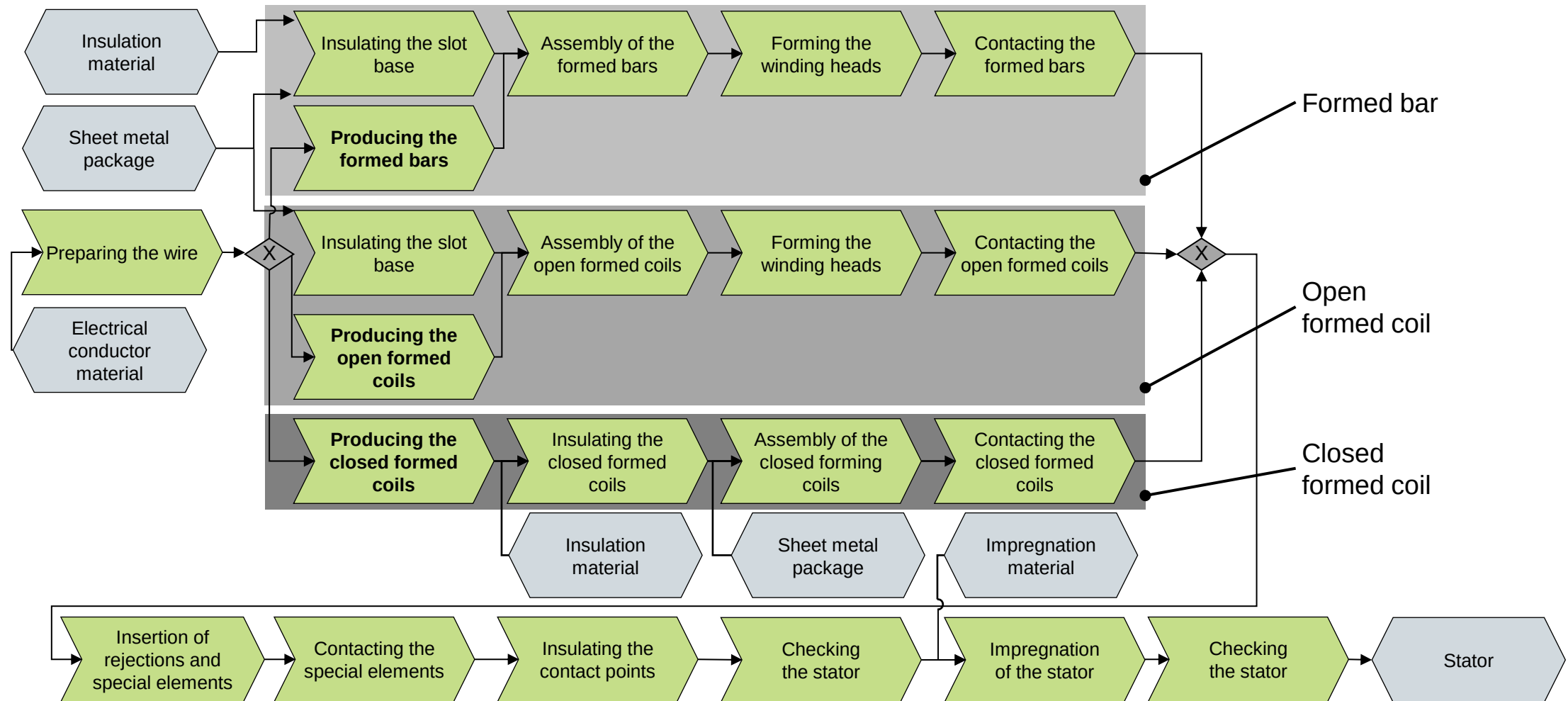


## The complexity of open formed coils, closed formed coils and formed bars differs significantly.

| <ul style="list-style-type: none"> <li>Coil fully formed and insulated outside the machine</li> <li>No significant deformation when assembled in slot</li> <li>Differentiation by shape</li> <li>varying complexity in the production of windings</li> </ul> |                                                                                                                       |                                                                                     |                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                              | Formed bar                                                                                                            | Open formed coil (e.g. hairpin)                                                     | Closed formed coil                                                                                                              |
| Geometry                                                                                                                                                                                                                                                     |                                     |  |                                              |
| Forms of execution                                                                                                                                                                                                                                           | <br>Semi-formed bar      Formed bar | --                                                                                  | <br>Needle coil      Oval coil      Fishbone |
| Slot insulation                                                                                                                                                                                                                                              | Sheet metal package / forming coil                                                                                    | Sheet metal package / forming coil                                                  | Formed coil                                                                                                                     |
| Mounting direction                                                                                                                                                                                                                                           | axial / radial                                                                                                        | axial / radial                                                                      | radial                                                                                                                          |
| Design complexity                                                                                                                                                                                                                                            |                                   |                                                                                     |                                                                                                                                 |
| Assembly complexity                                                                                                                                                                                                                                          |                                   |                                                                                     |                                                                                                                                 |
| Number of contact points                                                                                                                                                                                                                                     |                                   |                                                                                     |                                                                                                                                 |

The process chains differ significantly depending on whether formed bars, open formed coils or closed formed coils are used.

Slide from LU7



Open shaped coils made of rectangular enameled copper wire are suitable for axial mounting and are increasingly being used for smaller generators (alternators) and traction drives.

Slide from LU7



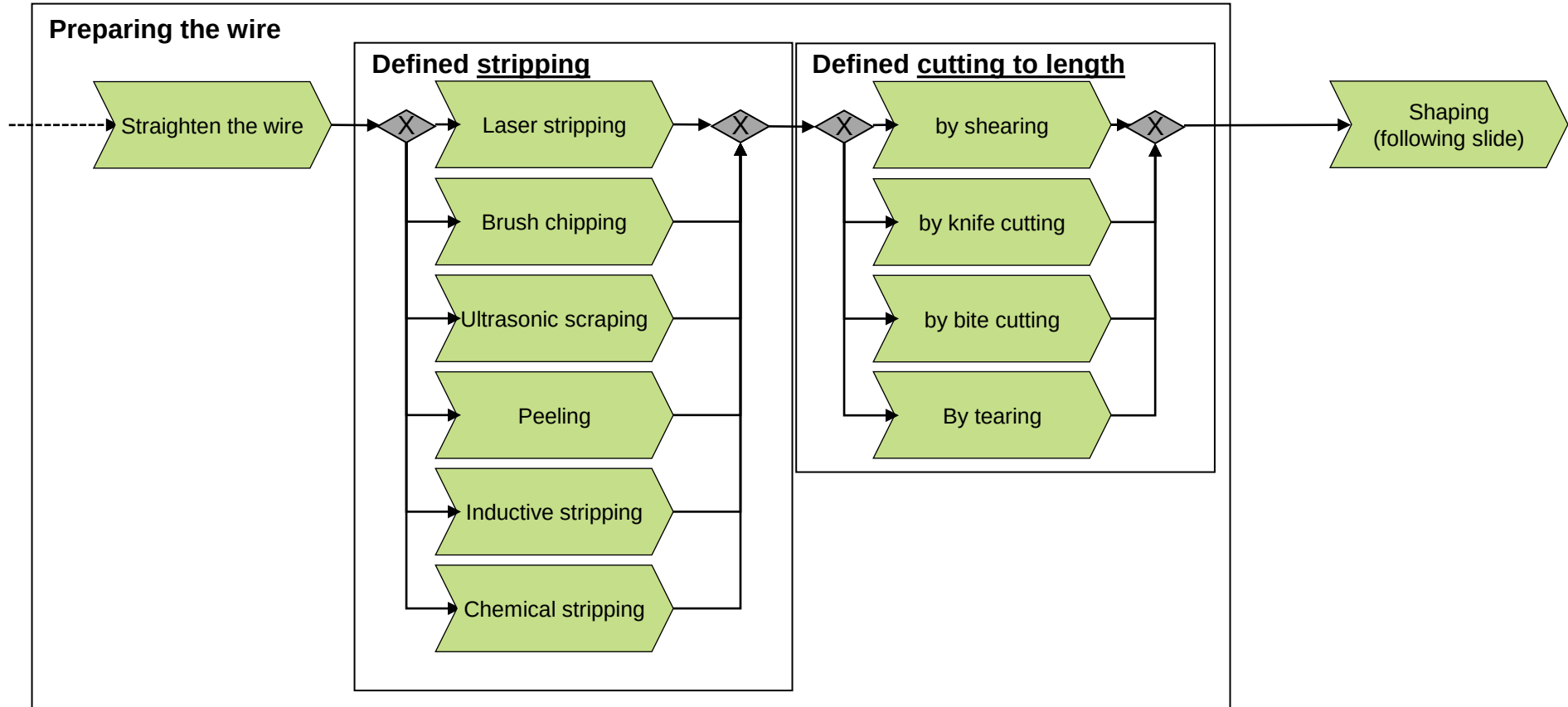
Video: GM

### Use of open form coils in medium and low power electrical machines

- Use of enamel-insulated copper conductors
- Axial mounting enabled by insertion
  - Direct insertion of groove base insulation in sheet metal package
  - Assembly of open molded coils even with closed grooves
- Topping required for correct positioning of the conductor ends after installation and before contacting
  - If necessary, bend the ends of the ladder radially ("teasing") to create space between levels
  - Cabinets for joining conductor ends to be connected ("twists")

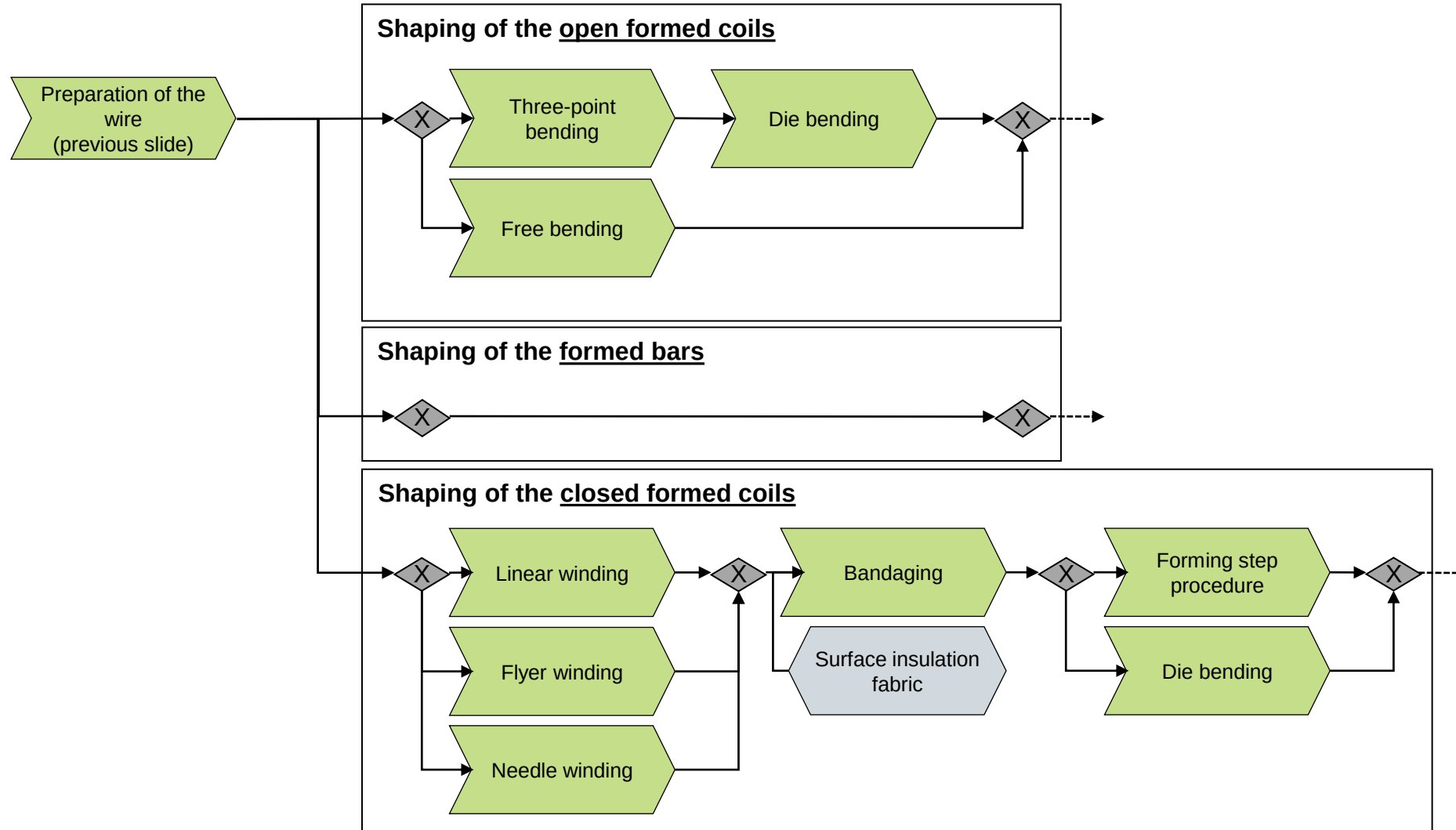
There is no difference in the preparation of the wire for the production of formed bars and open/closed formed coils.

Slide from LU7



There is no difference in the preparation of the wire for the production of formed bars and open/closed formed coils.

Slide from LU7



Open formed coils can be manufactured using free-form bending or tool-based processes.

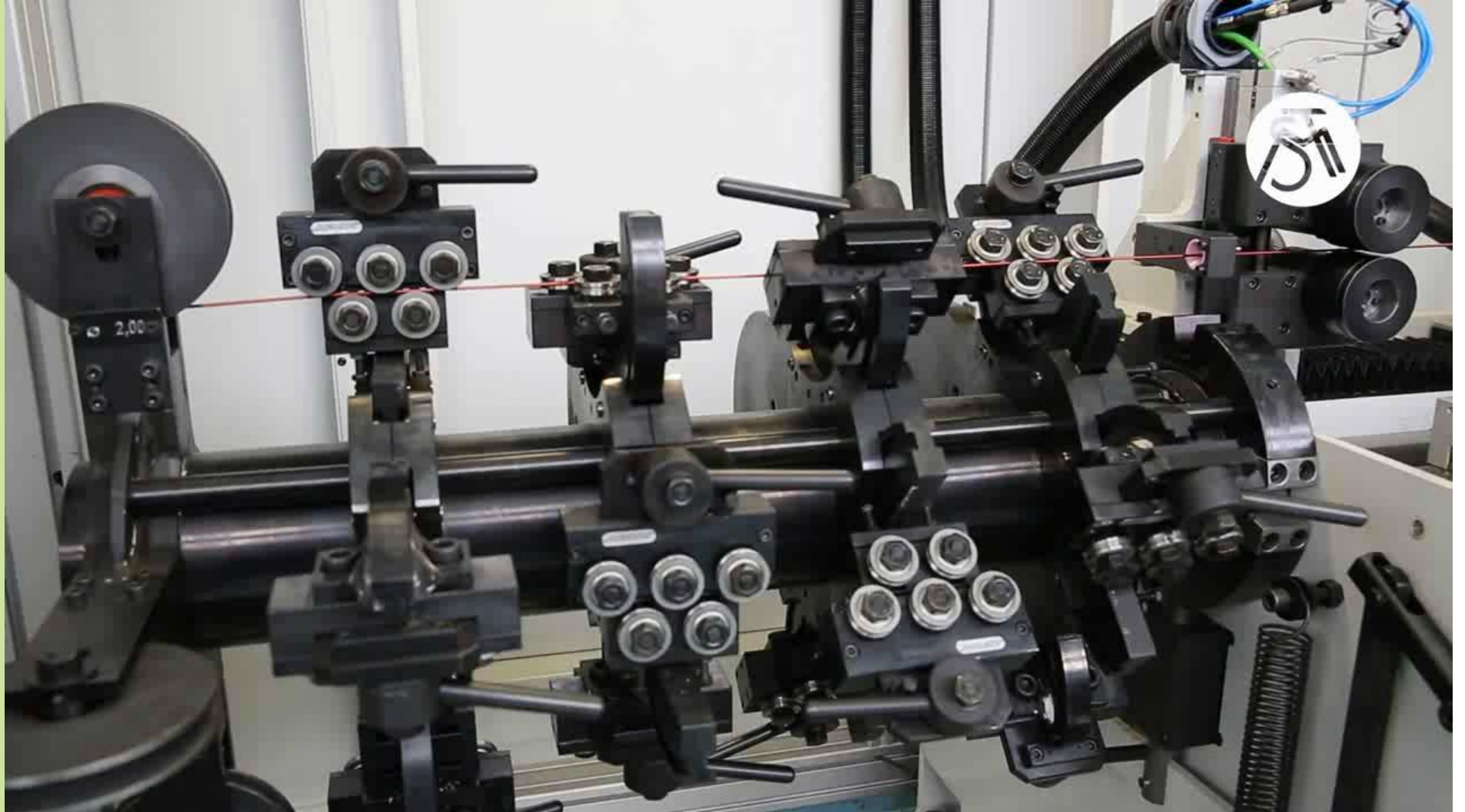
Slide from LU7

## Process

Process chain for the production of stators with open form coil windings

Video:  
SMT

[youtube.com/watch?v=X5VPRWnuQLQ](https://youtube.com/watch?v=X5VPRWnuQLQ)



The production and assembly of formed bars is carried out in simple processes, but a high number of contact points must be realized.

Slide from LU7

## Process

Process chain for the production of stators with shaped rod winding

Video:  
Continental

[youtube.com/watch?v=ft04LbFXR7E](https://youtube.com/watch?v=ft04LbFXR7E)



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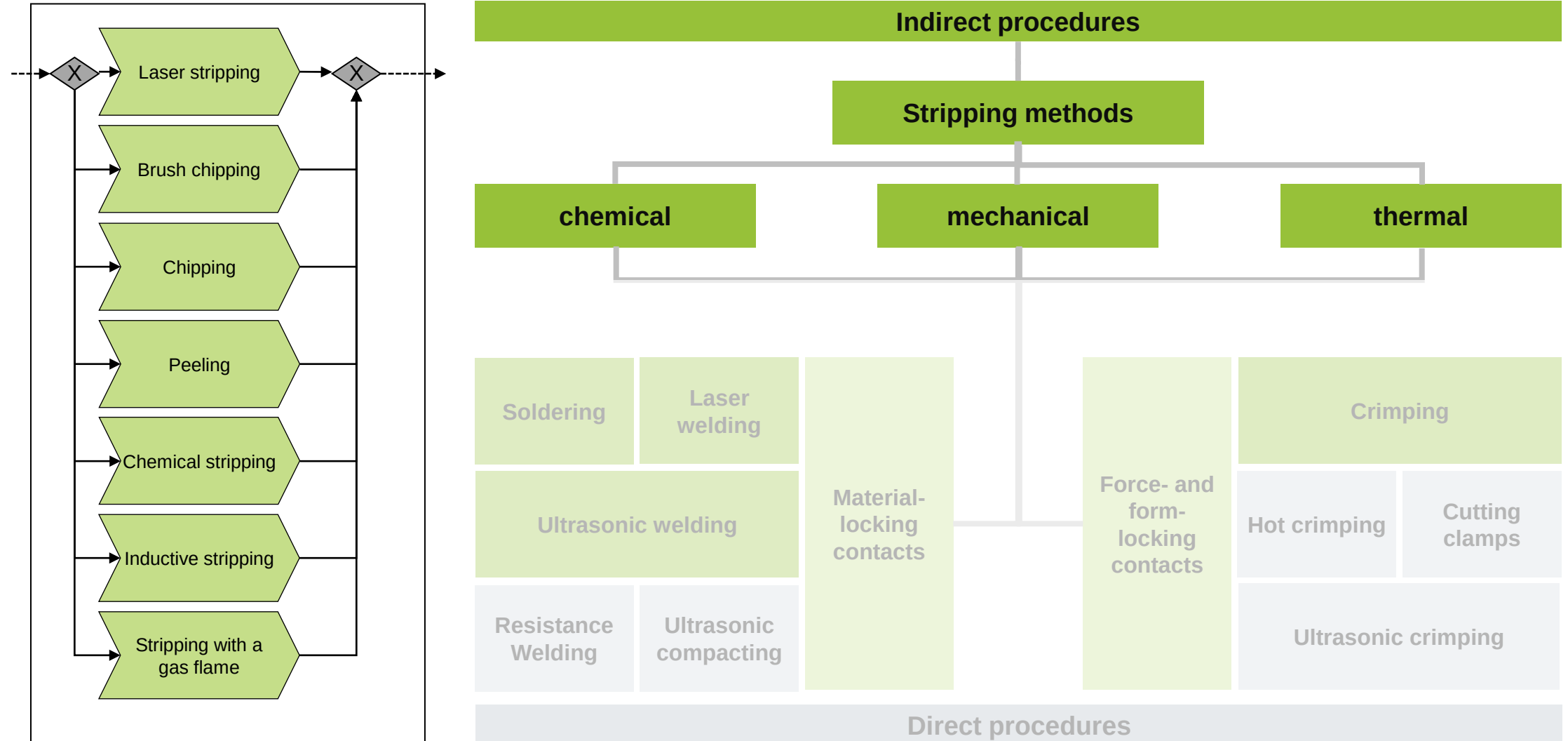
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Picture: VW

In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.

Slide from LU8



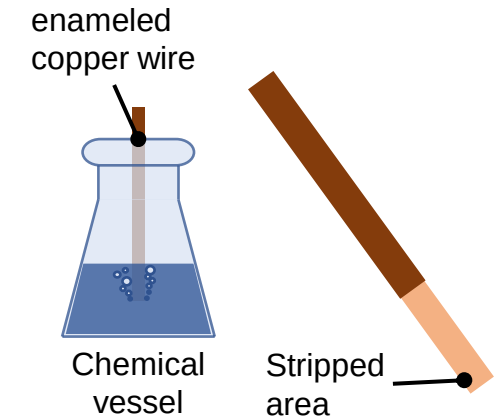
## The insulating layer can be removed from the enameled copper wire in different ways.

Slide from LU8

### Chemical stripping

Insulated conductor ends are exposed to the chemical "stripper" for a defined exposure time and then cleaned

- Chemical strippers have a strong dissolving power for Novolak- and epoxy-based enamels and polyimides
- At higher application temperatures of approx. 75 °C, also strongly cross-linked enamels
- Consist mainly of solvents (e.g. dimethyl sulphoxide) and bases (tetramethylammonium hydroxide)



### Advantages

- Comparatively cost-effective and low effort
- Application in prototype construction
- Reproducible results

### Disadvantages

- Very long process times (several hours depending on the degree of insulation and material)
- Handling of substances that are harmful to health
- Heating the "stripper" to temperatures almost below the flash point
- Not suitable for wire bundles, stranded wires or thermoplastic insulation (e.g. PEEK)

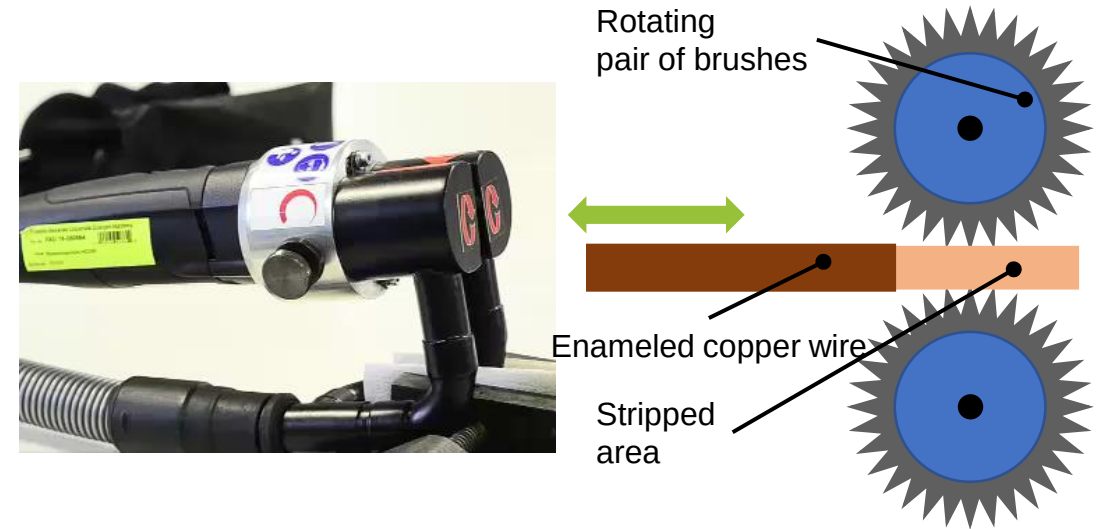
## The insulating layer can be removed from the enameled copper wire in different ways.

Slide from LU8

### Rotating brushes

The enameled copper wire is inserted between two opposing rotating steel brushes, whereby the insulation layer is removed and vacuumed off.

- Rotating steel brushes are arranged opposite each other
- Distance is adapted to the wire diameter
- Inserting the insulated copper wire
- Removing the insulation due to friction



### Advantages

- Low costs
- Simple process control
- Easy to use
- Suitable for various cross-sections
- Applicable for small quantities
- State of the art

### Disadvantages

- Fluctuating removal quality and tool wear
- Damage to the wire due to mechanical forces
- Danger of insulation residues
- Undefined transition between stripped and insulated area
- Damage to the conductor surface by brush wires
- Not suitable for melting thermoplastics

## The insulating layer can be removed from the enameled copper wire in different ways.

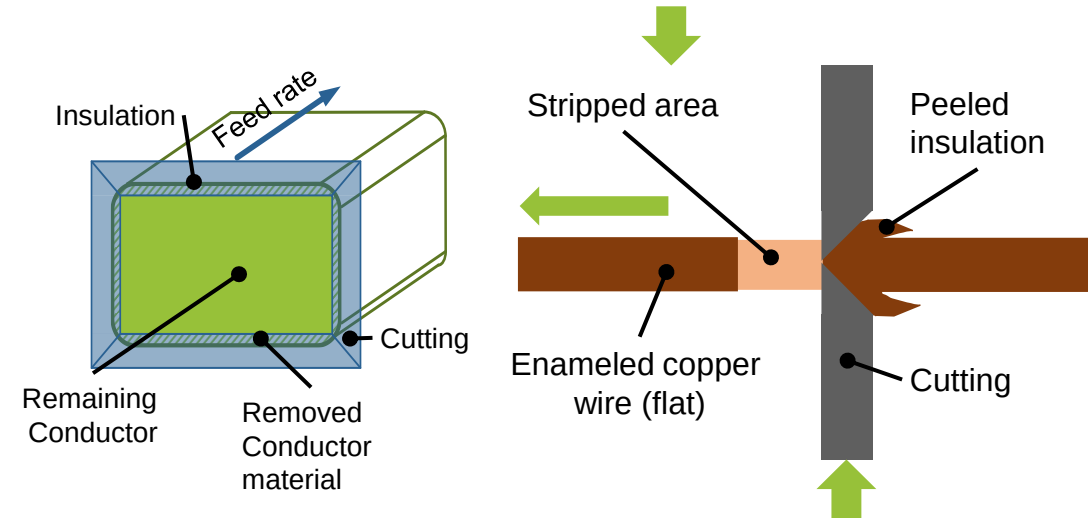
### Peeling

Blades cut into the copper material in a rectangular geometry and realize a feed movement along the conductor, whereby the insulation is removed without leaving any residue.

- Kinematic structure usually consisting of four cutting edges
- Cutting edges penetrate the insulation and cut into the conductor material, linear feed movement of the cutting edges along the conductor or of the conductor along the cutting edges leads to peeling of the insulation
- Due to the shape of the flat copper conductors (radii at edges), removal of conductor material cannot be avoided

### Advantages

- Easy to use
- Defined cutting line
- Defined transition area
- Simple process principle
- Independent of insulation type
- residue-free removal



### Disadvantages

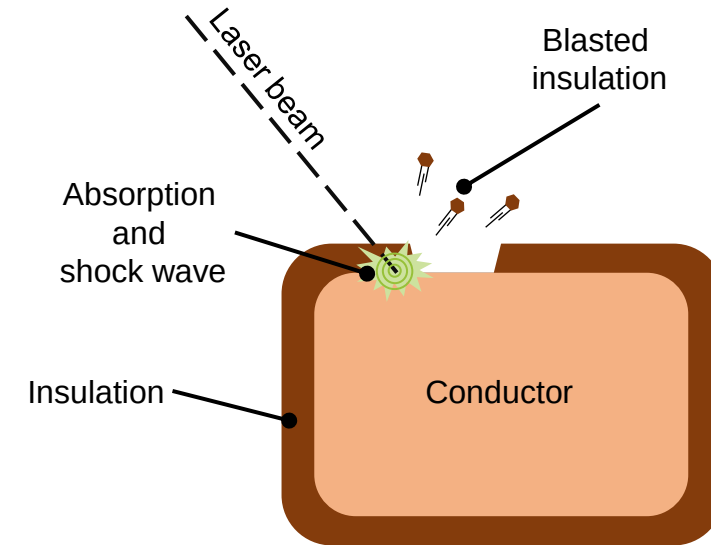
- Removal of conductor material by penetration of the cutting edges into the copper material
- Depth must be adapted to the edge radius
- Tool wear
- Challenging positioning of the cutting edges

## The insulating layer can be removed from the enameled copper wire in different ways.

### Laser: Near infrared radiation

Near infrared radiation is partially absorbed by the surface of the copper conductor after passing through the insulation. The pulsed laser power causes the copper material to heat up abruptly and expand without melting.

- Use of solid-state lasers (e.g. Nd:YAG laser,  $\lambda \sim 1,064 \text{ nm}$ ) in pulsed operation
- Shock waves are generated which cause the insulation to flake off



### Advantages

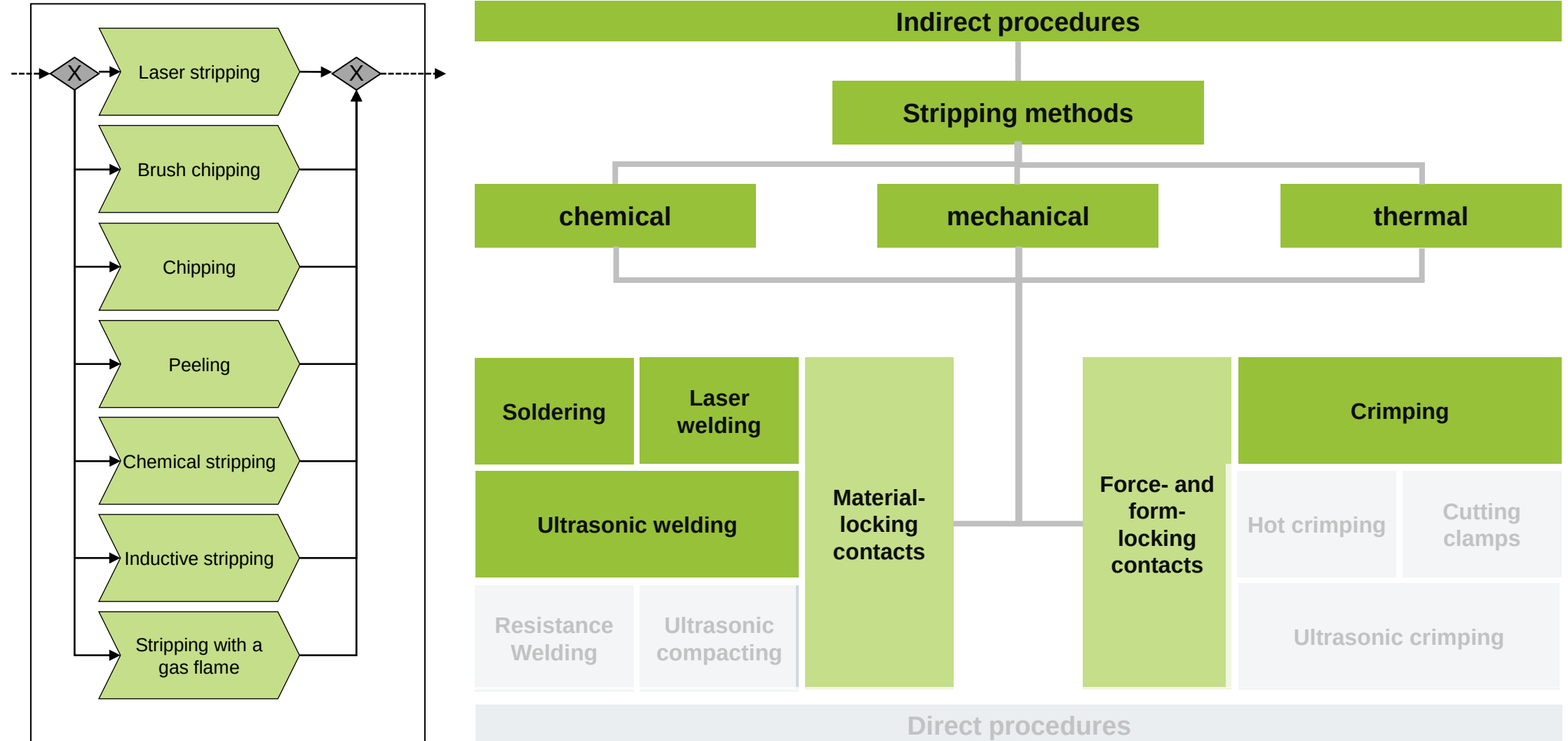
- Non-contact method
- Free of residues
- Defined cutting line
- High removal rate

### Disadvantages

- Complex process
- High investment costs

In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.

Slide from LU8



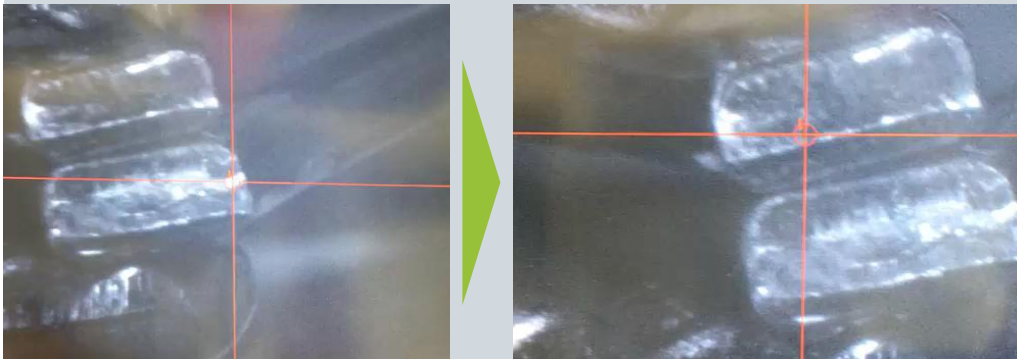
The laser welding process is mainly used for molded coil windings such as hairpins.

Slide from LU8

### Laser beam welding

- Use of high-brilliance solid-state lasers (e.g. disc lasers) in the infrared wavelength range
- Use of high welding power to form a keyhole despite high reflection of the laser radiation at the start of the process
- Melting of the contact partners and formation of a material locking bond
- Mainly used for flat wires and single wires
- Further optimization of the laser beam welding process through the use of green / blue lasers

Videos: Keyence



### Advantages

- Extremely short process times
- Scanner-based beam guidance
- High process stability
- Realization of high welding depths
- Contactless process

### Disadvantages

- High system investment
- Sensitive to insulation residues and positioning errors

Laser beam welding of preformed coil windings shows potential for producing the large number of contact points on preformed coil windings.

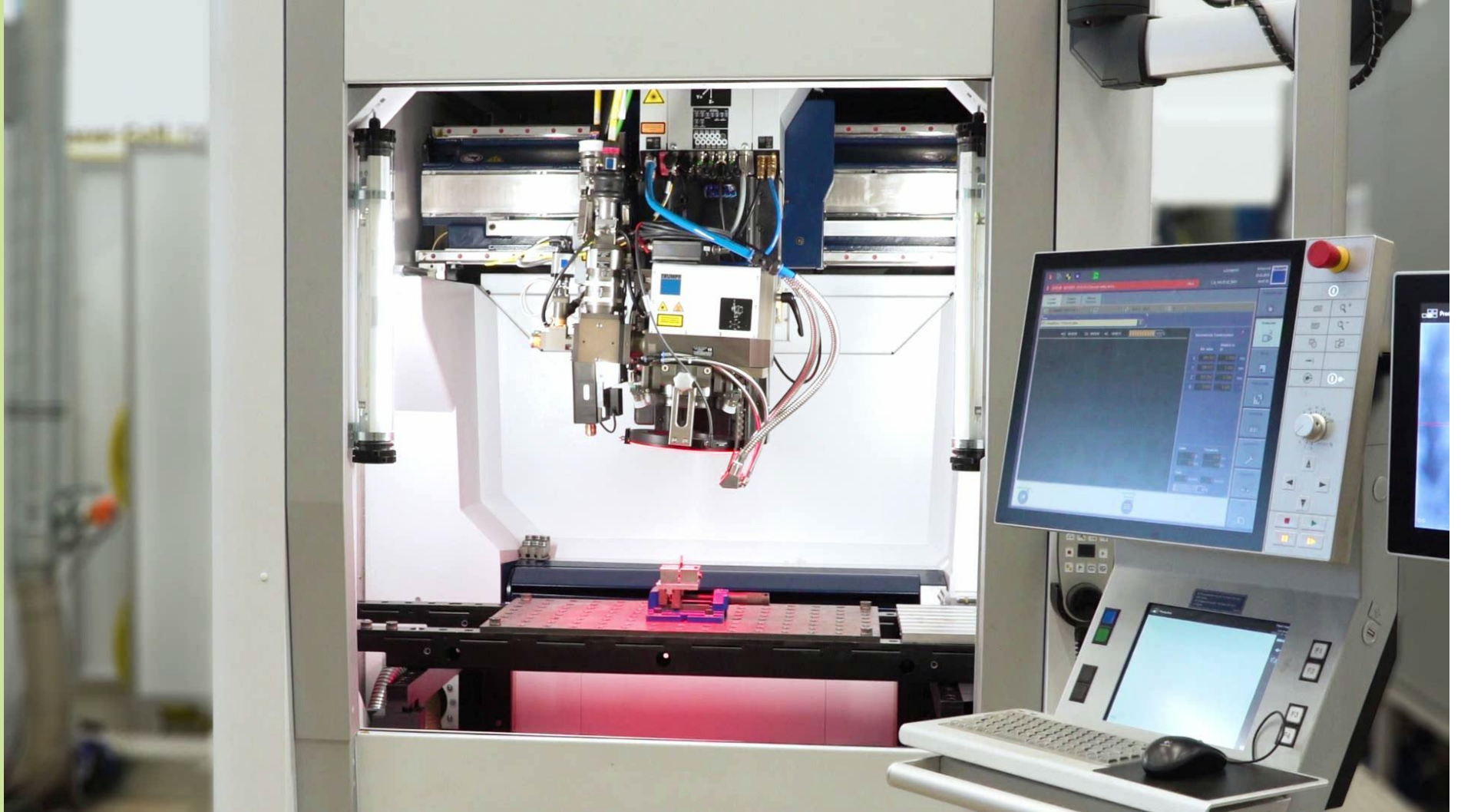
Slide from LU7

## Process

Contacting flat conductor-based  
shaped coil windings using the laser  
welding process

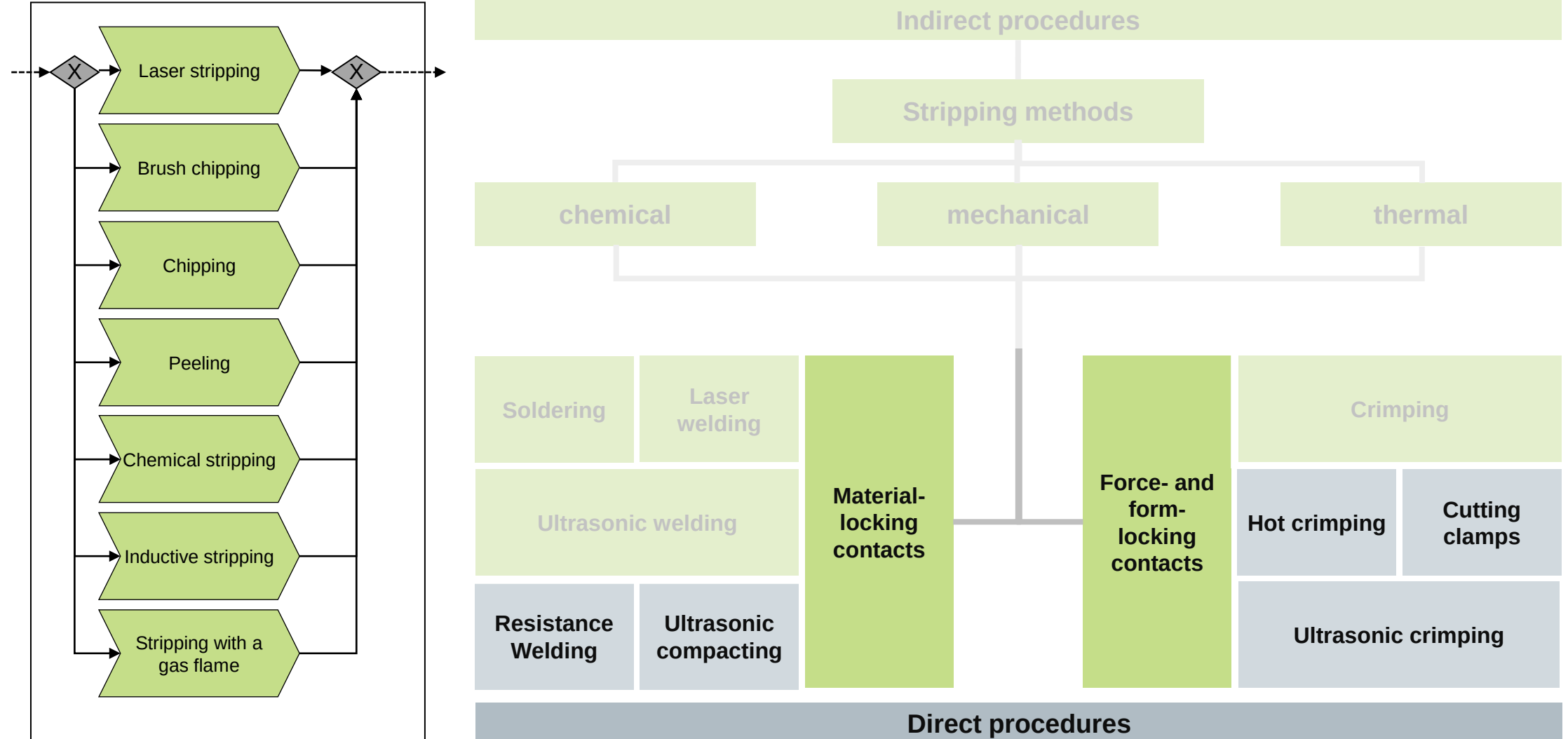
Video:  
Trumpf

[youtube.com/watch?  
v=kW0zPOvUY2M](https://youtube.com/watch?v=kW0zPOvUY2M)



In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.

Slide from LU8



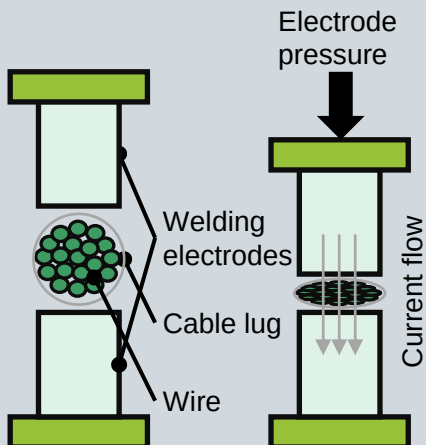
## The hot crimping process reflects the current industrial standard for contacting enameled wire bundles.

Slide from LU8

### Hot crimping

Hot crimping is a type of resistance welding in which a conductive connection part (a cable lug, terminal, or a simple round sleeve) is used.

- The heat generated by the flowing current (Joule heating) causes the insulation to evaporate and the wires to bond together
- By integrating the process step of stripping, productivity in the production of electrical connections is increased



Video: FAPS



### Advantages

- Stripping and contacting of enameled wire bundles
- Good automation and process control
- Good mechanical and electrical connection quality
- Short process times

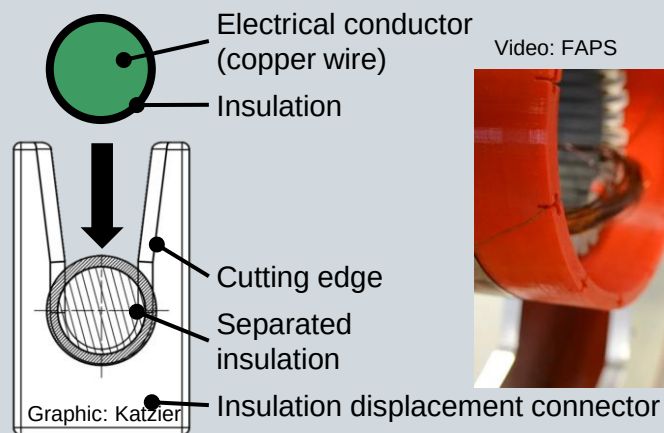
### Disadvantages

- High tool wear (depending on electrode material)
- Combustion residues from the insulation
- High process energy requirement

The insulation displacement method is almost never used in the production of electrical machines due to its susceptibility to vibration.

### Insulation displacement method

- Mechanical stripping of the enameled wires during the insertion process on the sharp-edged legs of the insulation displacement connectors
- Contact is made by applying pressure (force-fit) to insulation displacement connector
- Insulation displacement connectors have mechanical pretension
- Insulation displacement connectors must be matched to the wire cross-section



#### Advantages

- Possibility of full automation
- Low process costs

#### Disadvantages

- No multi-wire connections (max. two wires of the same diameter)
- Lack of process reliability for the automotive sector
- No strain relief
- Sensitive to vibration

In the contacting process step, the free conductor ends are electrically and mechanically connected.

### Contacting technologies

| Technology            | Automation capability | Flexibility | Tool wear | Process reliability | Accessibility | Cycle time |
|-----------------------|-----------------------|-------------|-----------|---------------------|---------------|------------|
| Resistance welding    | +                     | 0           | --        | 0                   | 0             | +          |
| Laser welding         | ++                    | +           | ++        | ++                  | ++            | ++         |
| Ultrasonic welding    | +                     | 0           | +         | +                   | -             | ++         |
| Crimping              | ++                    | -           | +         | +                   | 0             | ++         |
| Hot crimping          | +                     | +           | --        | 0                   | -             | 0          |
| Ultrasonic crimping   | +                     | +           | +         | +                   | -             | 0          |
| Ultrasonic compacting | ++                    | 0           | +         | +                   | --            | ++         |
| Soldering (hard/soft) | --                    | ++          | ++        | -                   | ++            | 0          |
| Cutting clamps        | +                     | +           | ++        | 0                   | +             | ++         |

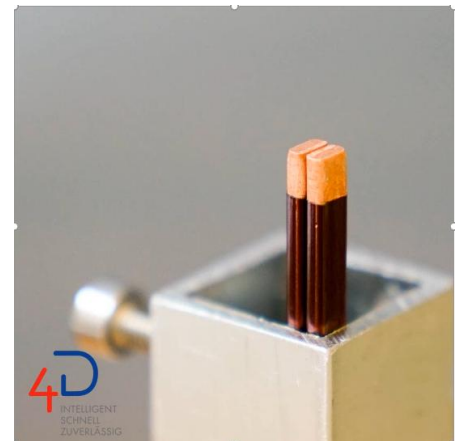


### Contacting of forming coils - forming rod windings

- Number of contact points
- Cycle time
- Off-peak times
- Process reliability
- Welding spatter
- Mechanical load
- Heat input
- Position tolerances
- Accessibility
- Connection cross-section
- Contact resistance
- Mechanical strength

### Potential of laser beam welding

- + Short cycle times
- + Beam guidance with scanner
- + Process stability
- + High welding depths possible
- + Contactless process
- Development of sputum



Video: 4D GmbH

The qualification of contact points can be carried out mechanically, electrically and with regard to long-term stability.

Slide from LU8

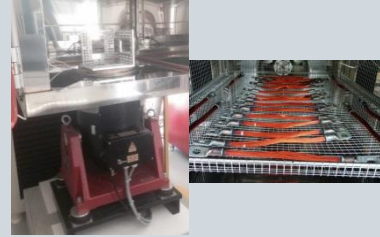
### Static mechanical testing

Pull-out tests



### Dynamic mechanical testing

- Temperature shock test
- Vibration test



### Corrosion testing

Salt spray test



There are several challenges in the qualification of contact points:

- Non-destructive testing
- Overfulfilment of standards
- Difficult to measure resistances
- Minor changes

Graphics & Videos: FAPS

### Imaging methods

- Microsections
- X-ray CT



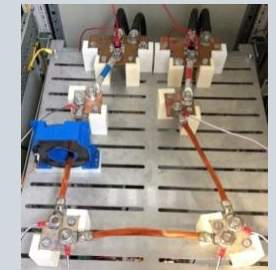
### Dynamic electrical testing

- Slow-Motion -Bending Test
- Online monitored dynamic resistance test



### Static electrical testing

- Measurement of contact resistances at high currents
- Nanoohmmeter



The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

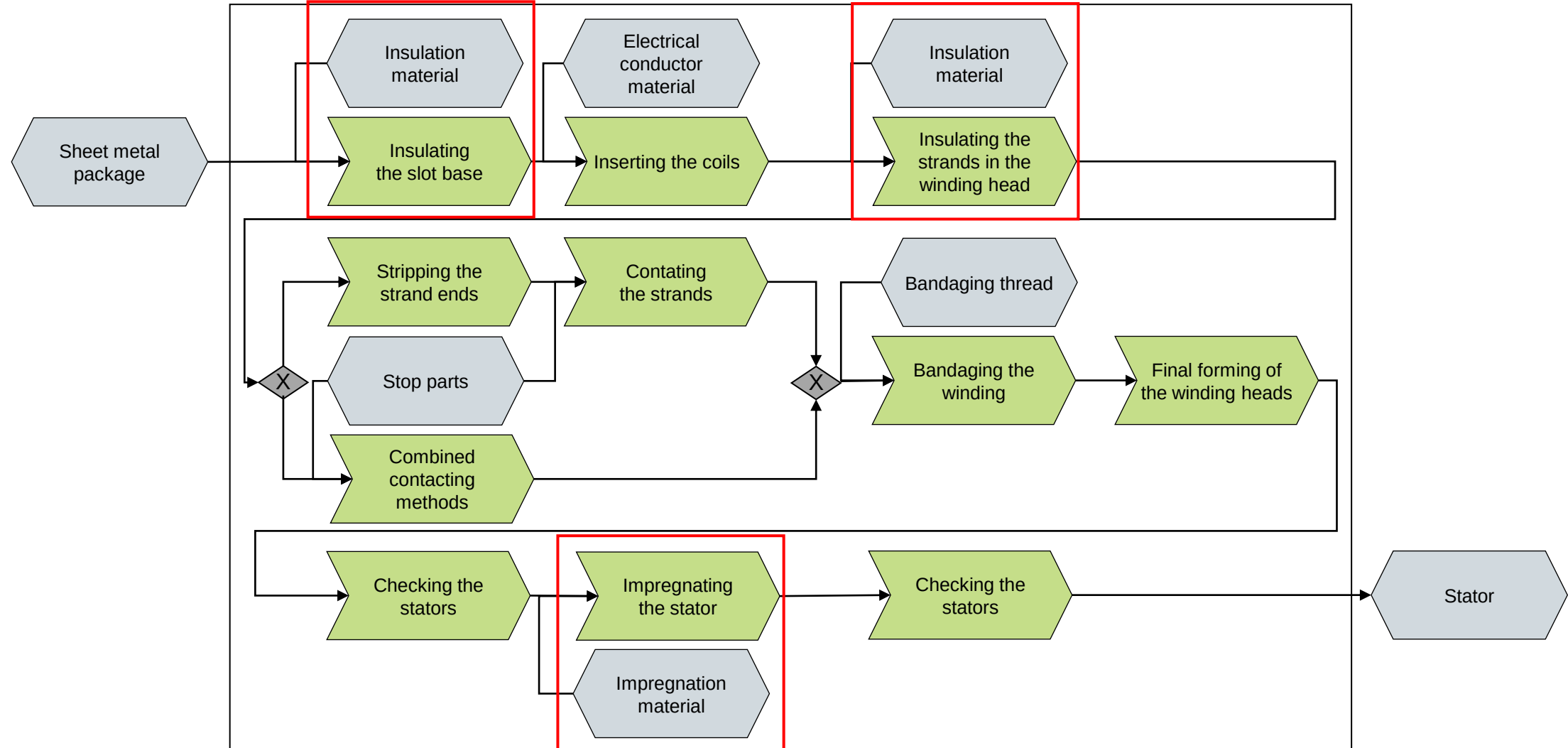
- Repetition of the production process chains
  - Higher-level production process chain of an electric motor
  - Production of the soft magnetic core
  - Insulation of the slot base
  - Winding process
  - Electrical contacting
  - **Insulation systems**
  - Permanently excited rotors
  - Mechanical components of electric motors
- Process chain analysis based on practical examples



Picture: VW

# The production chain for the manufacture of a stator with a distributed winding of round wire essentially consists of nine steps.

Slide from LU9



Insulators are primarily classified on the basis of their electrical properties.

Slide from LU9

### Contact resistance

- The resistance to the flow of current through a material
- often also simply "(electrical) resistance"
- has the unit Ohm  $1 \Omega = 1 \text{ V/A}$
- German: "Kontaktwiderstand"

### Surface resistance

- The resistance to the flow of current between two electrodes placed on the surface of an insulator
- has the unit Ohm  $1 \Omega = 1 \text{ V/A}$
- German: "Oberflächenwiderstand"

### Tracking resistance

- Resistance of an insulator to currents that form on the surface
- is determined with CTI value (voltage at which a leakage current occurs)
- German: "Kriechstromfestigkeit"

### Dielectric strength

- Maximum electric field strength that may prevail in the material without a voltage breakdown (electric arc) occurring
- Specification in kV/mm
- German: "Durchschlagsfestigkeit"

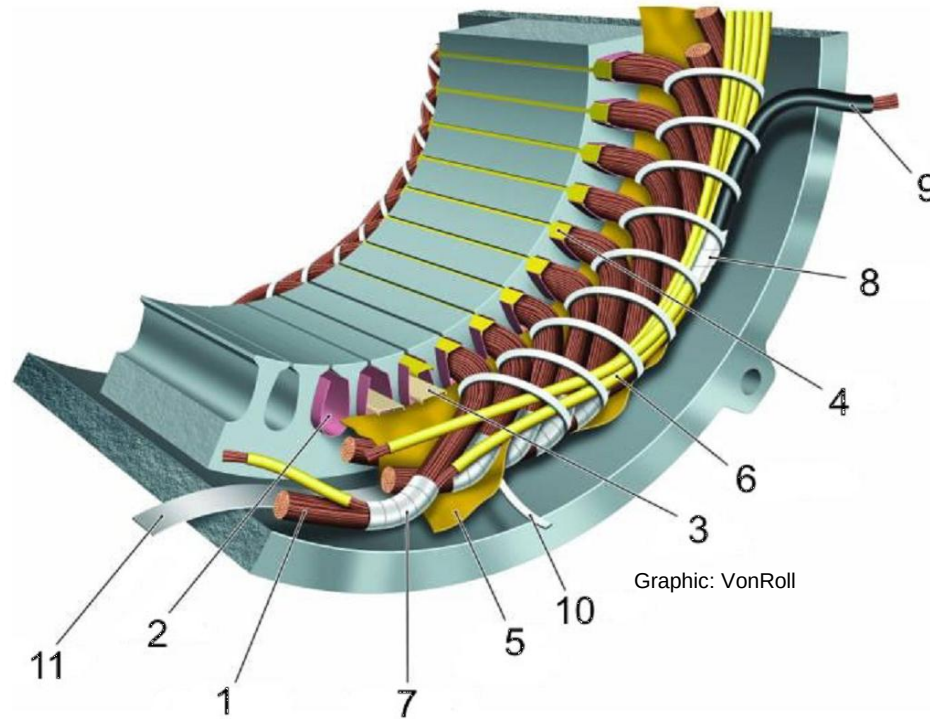


Graphic: rm-prüftechnik

- Testing of substances and insulated stators is carried out using dedicated measuring methods and devices
- Test methods are often destructive and damage the insulation

Insulating materials are applied to the electrical machine in a variety of forms and materials.

### Insulation types and variants in low-voltage machines < 1 kV



1. Wire insulation
  2. Groove base insulation
  3. Groove separation
  4. Cover slide
  5. Phase separation
  6. Cable insulation
  7. Bandage insulation
  8. Bandage insulation
  9. Cable insulation
  10. Wrap-around head bandage
  11. Bandage insulation
- Without number: Impregnation

#### Primary insulation

Insulation layer that directly encloses the electrical conductor (wire enamel, sheet metal coating, bandages)

#### Secondary insulation

Additional insulation materials that are added during the assembly process (impregnating resin, insulation paper, groove separating slider)

After the contacting process, the exposed contact points must be isolated.

Slide from LU7

Contacting



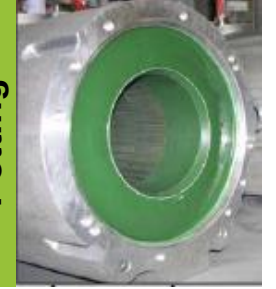
#### Tasks

- Isolation Contact points to each other
- Protection against moisture and dirt

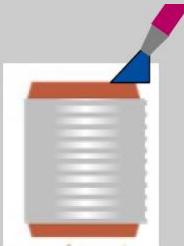
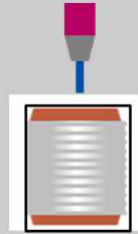
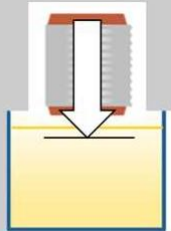
Coating



Potting



### Technologies for insulating the contact points of forming coil/forming rod windings

|                      | Spraying                                                                           | Drizzle                                                                              | Current UV                                                                           | Potting                                                                              | Diving                                                                               |
|----------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Insulation materials | Resins                                                                             | Resins                                                                               | Resins                                                                               | Resins                                                                               | Powder or resins                                                                     |
| Principle sketch     |  |  |  |  |  |
| Advantages           | Simple procedure                                                                   | Low soiling Stator                                                                   | High dielectric strength                                                             | High thermal conductivity                                                            | Simple procedure                                                                     |
| Disadvantages        | Thin insulation layer on edges                                                     | Good accessibility required                                                          | Contamination of complete stator                                                     | High mass increase                                                                   | Low reproducibility                                                                  |

Source/Pictures: ELANTAS

In the industry, immersion roller burnishing and trickling are widely used impregnation processes.

### Dip rolling



#### Advantages

- Even resin distribution
- Less material loss (dripping)
- Suitable for the automated production process
- Inexpensive and simple process

Videos: bdtronic

### Drizzle



#### Advantages

- Good material utilization, low drip losses
- High automation capability
- Short cycle times and high quantities possible
- Can be combined with electric heating or oven curing

The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

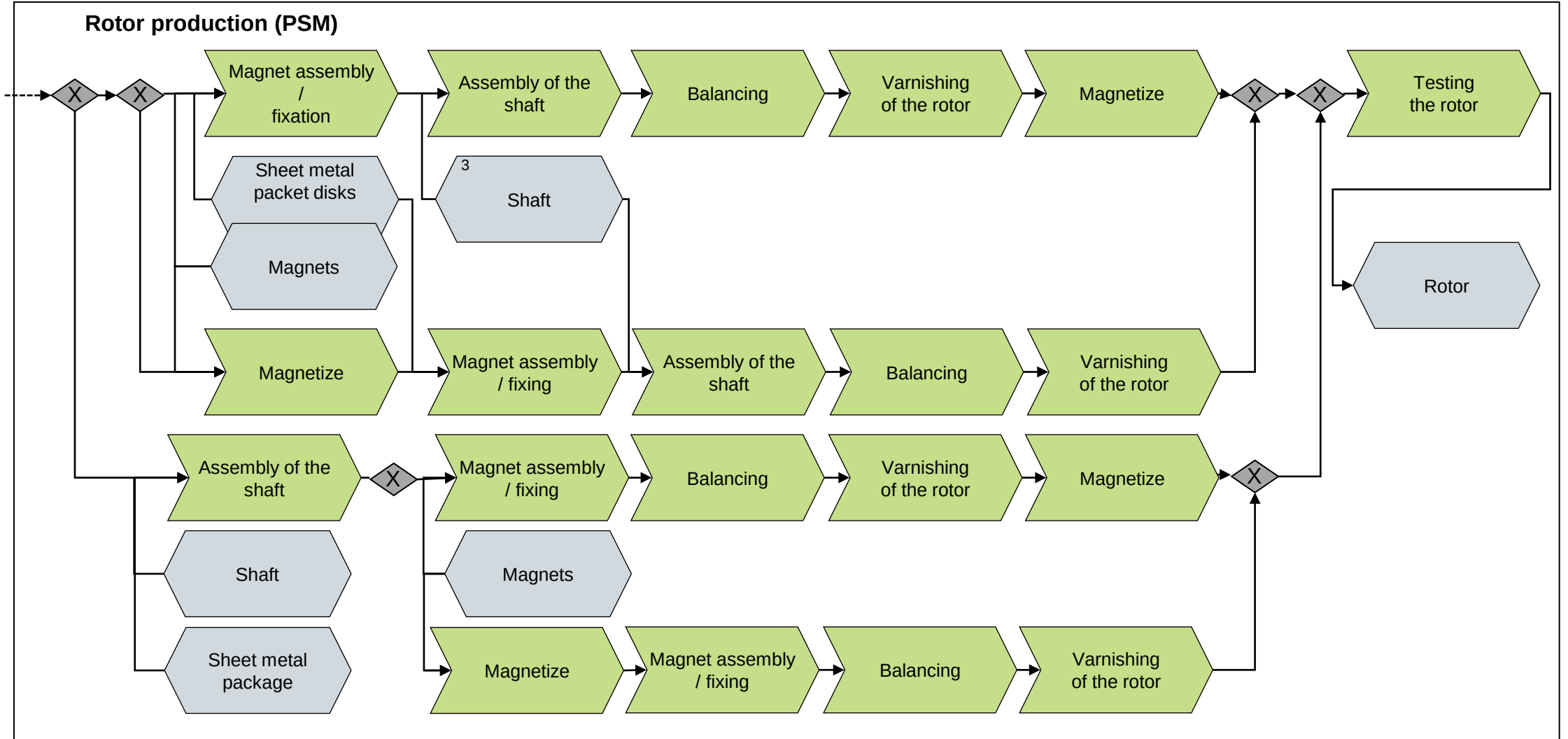
- Repetition of the production process chains
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  - Production of the soft magnetic core
  - Insulation of the slot base
  - Winding process
  - Electrical contacting
  - Insulation systems
  - **Permanently excited rotors / Permanent Magnet Synchronous Machine**
  - Mechanical components of electric motors
- Process chain analysis based on practical examples



Picture: VW

# Various process steps are used to manufacture the rotors of a permanent magnet synchronous machine (PSM).

Slide from LU4



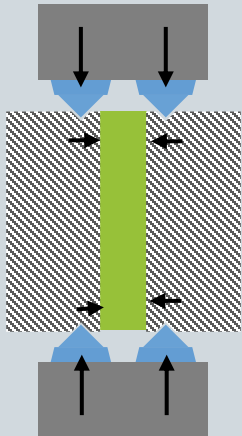
For the magnet mounting and fixing process steps, alternative joining technologies can be used.

Slide from LU4

### Caulking

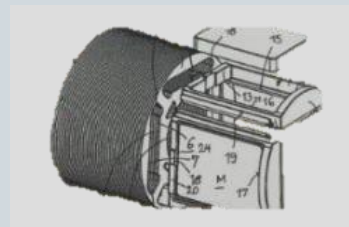
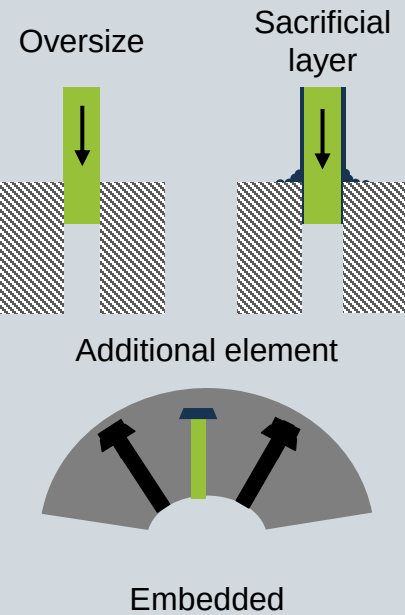
Force fit

Reshaping the package with a stamp



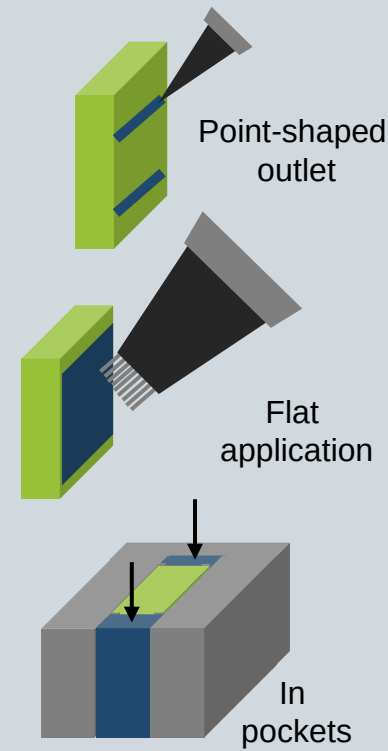
### Clamping / Pressing in

Form fit / Force fit



### Gluing

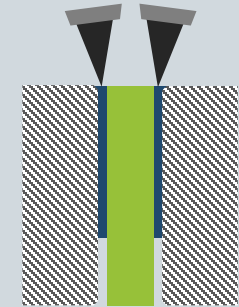
Material locking



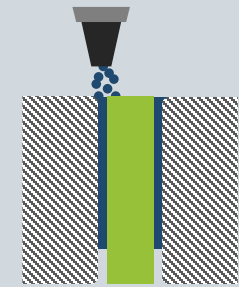
### Pouring

Material locking

Injection molding or compression molding



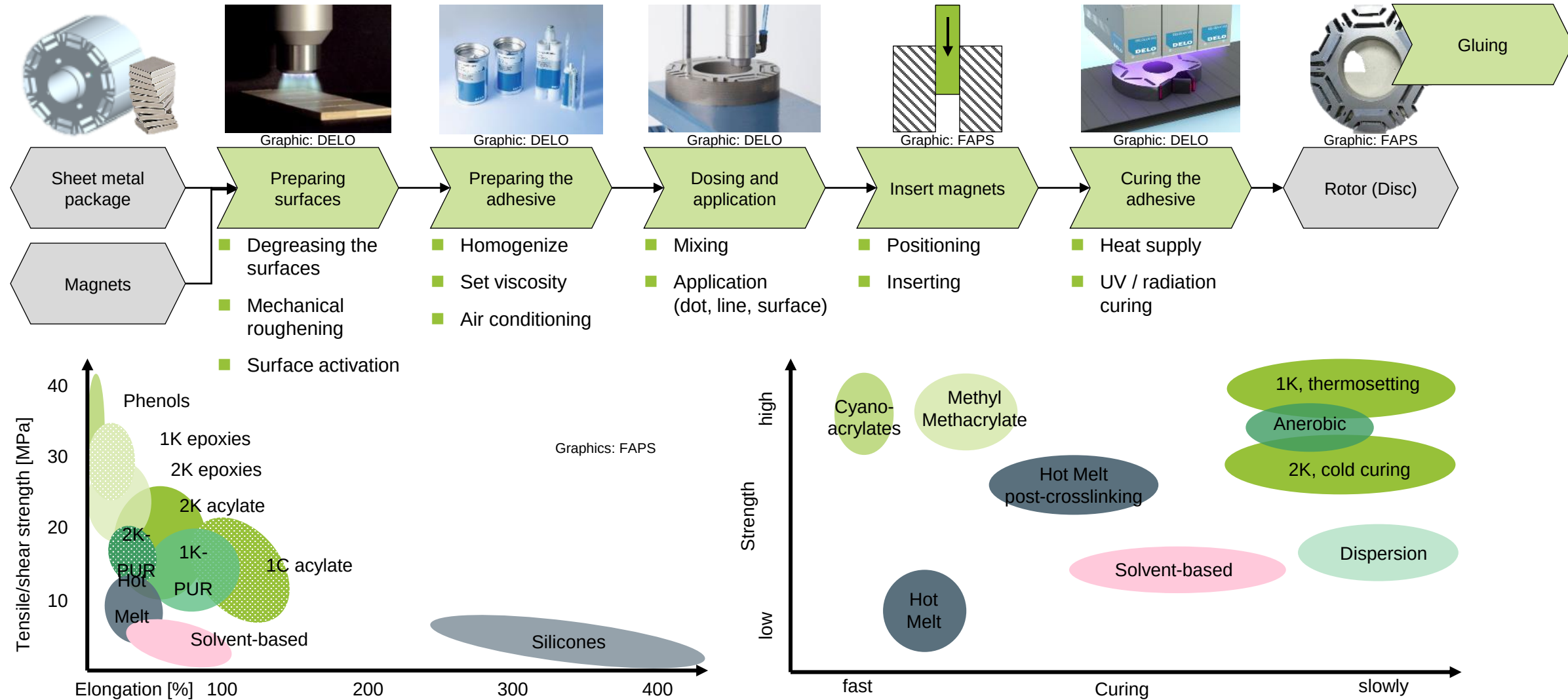
Drizzled



Magnet assembly

The bonding of magnets has established itself as a standard process and is therefore considered as an example.

Slide from LU4



The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

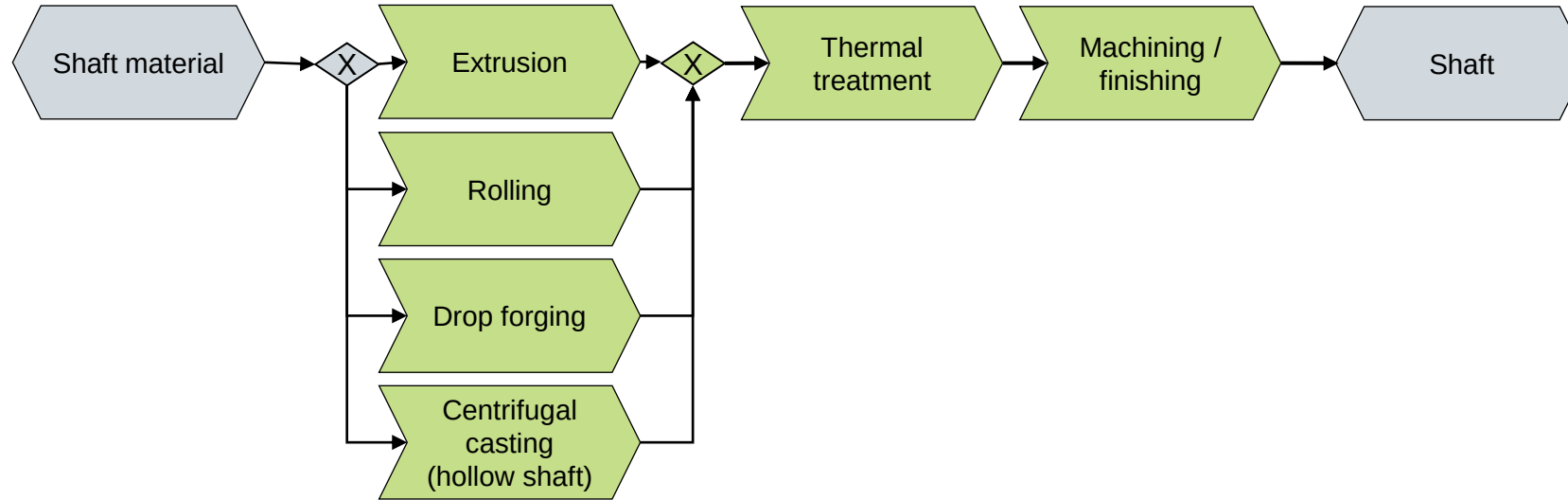
- Repetition of the production process chains
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  - Insulation systems
  - Permanently excited rotors
  - **Mechanical components of electric motors**
- Process chain analysis based on practical examples



Picture: VW

The shaft is first produced by casting, then by pressing, rolling, or forging and finally by machining.

Slide from VE10



Impact extrusion



Graphic: Schuler

Thermal treatment



Graphic: Hirschvogel

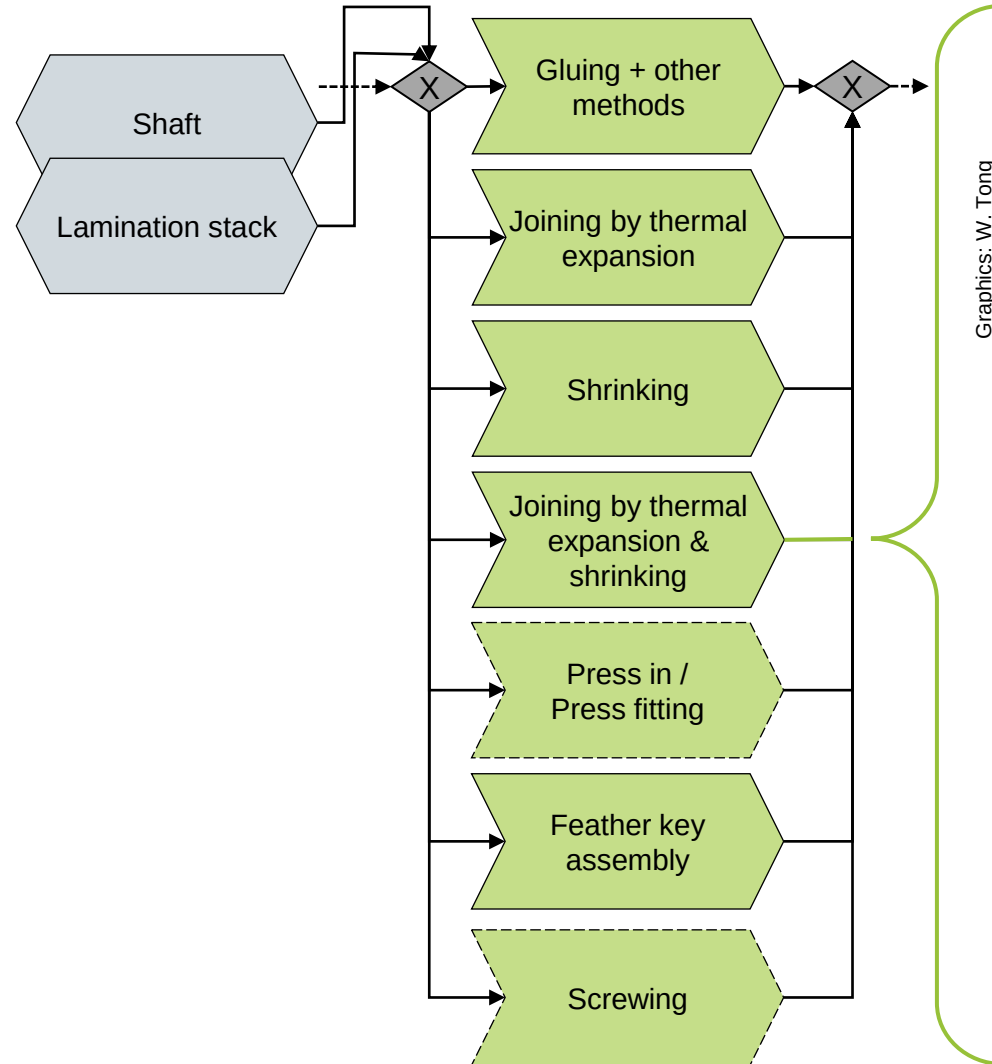
Machining / finishing



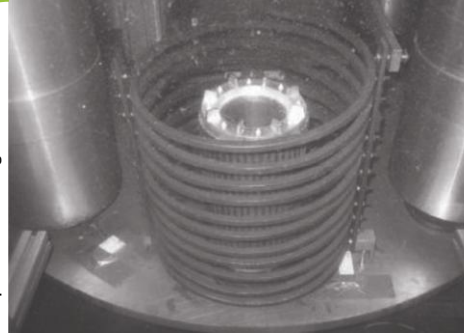
Graphic: Hirschvogel

# The joining of shaft and lamination by a thermal cross-press connection is the predominant method for traction drive applications.

Slide from LU10



Graphics: W. Tong



## 1<sup>st</sup> Step

Inductive heating of lamination



## 2<sup>nd</sup> Step

Subcooling the shaft with liquid nitrogen



## 3<sup>rd</sup> Step

Cooling / heating at room temperature

## Advantages of induction heating

- Time efficient
- Accurate
- Contactless
- Easy to control
- Less danger for workers compared to flames
- No heating of surrounding assemblies or environment

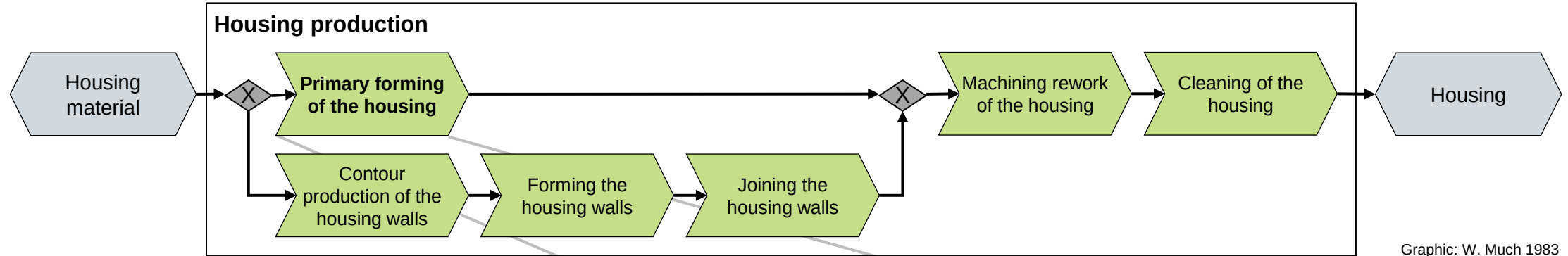
## Disadvantages of induction heating

- Electromagnetic interference of the environment possible
- Induction coil must be matched to lamination stack

**The combination of shrinking and sub cooling extends the process limits.**

The housing of an electric motor fulfills various tasks and is usually produced using conventional manufacturing processes.

Slide from LU10

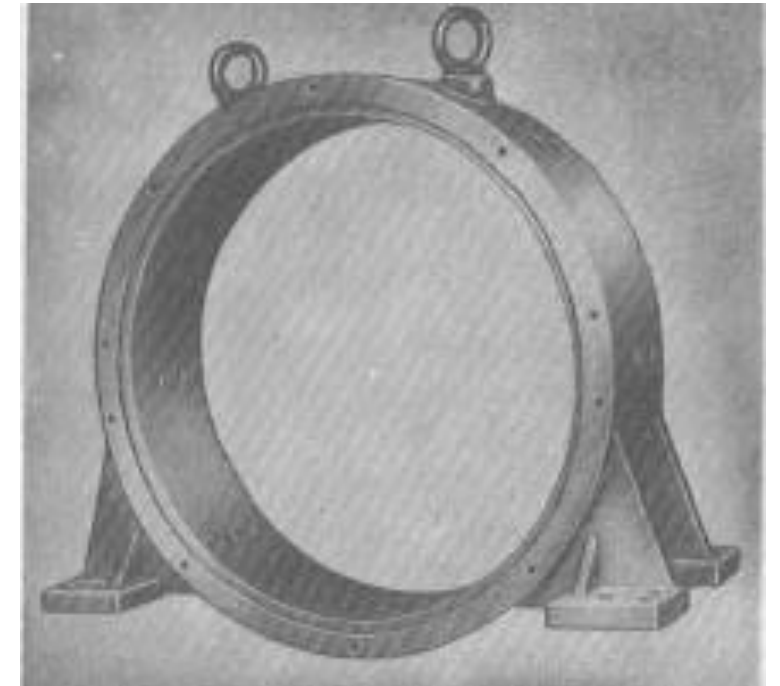
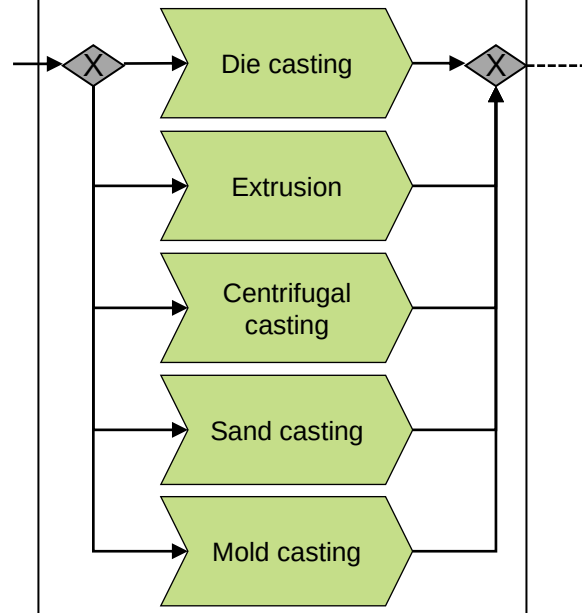


Graphic: W. Much 1983

### Tasks and requirements for the housing

- Torque support
- Adjusting and maintaining the coaxiality of the rotor and stator
- Securing and centering the bearing end shields
- Mounting for the power and signal connections
- Heat dissipation
- Protecting the active components from environmental influences
- Transport protection and positioning of the motor at the installation site

### Primary forming of the housing



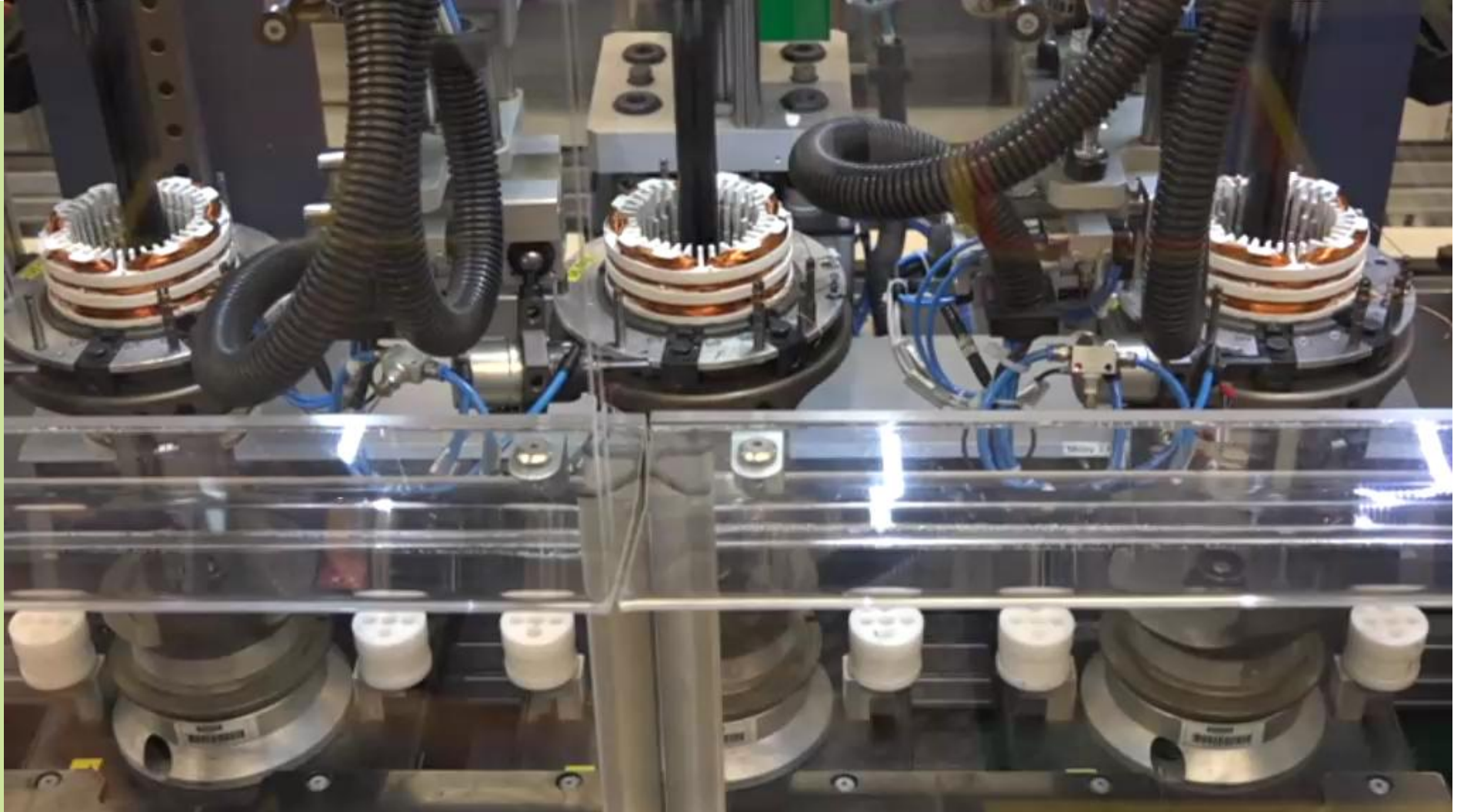
The motor housing is usually shrunk onto the stator, as this ensures a high degree of axially and good heat transfer can be ensured.

Slide from LU10

## Process

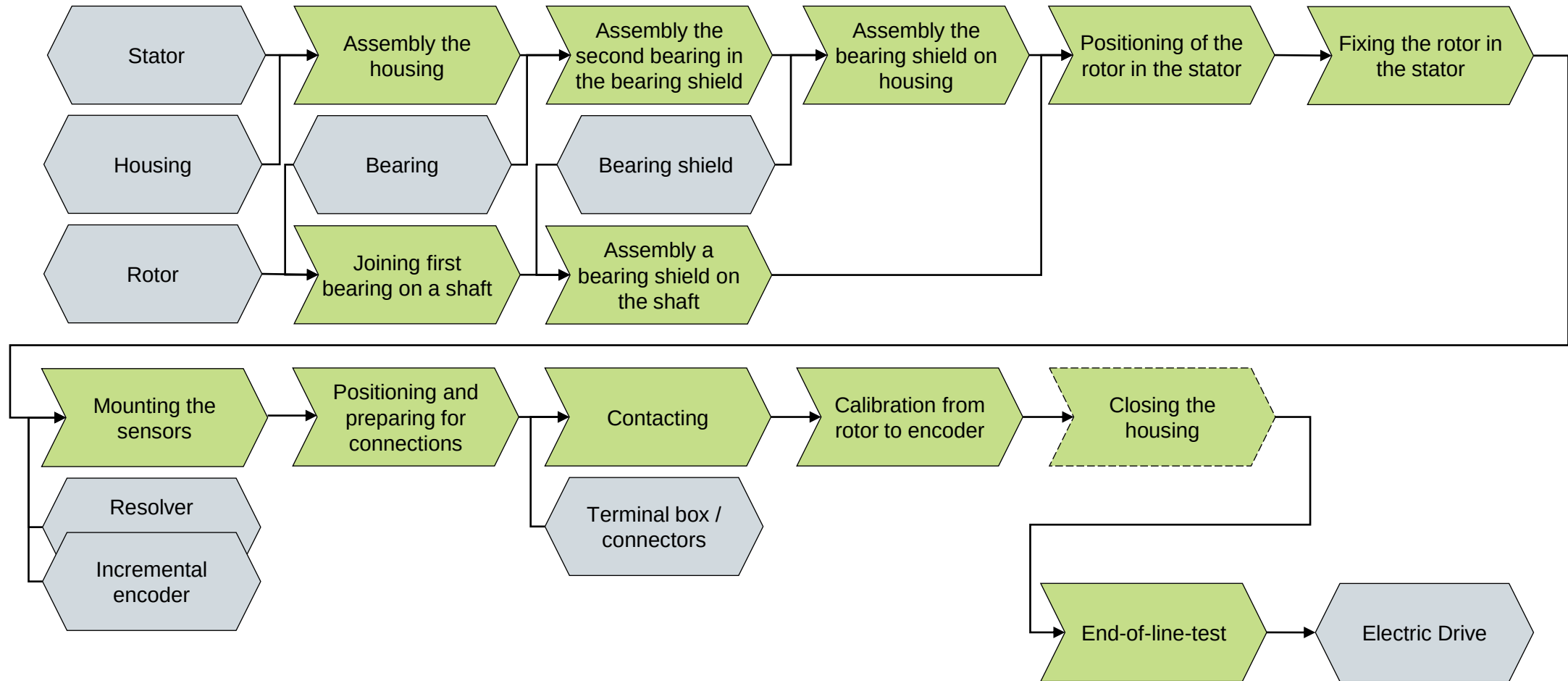
Shrinking the motor housing

Video: Siemens



The "marriage" of stator and rotor is followed by several assembly steps, depending on the design and functional scope of the motor.

Slide from LU10



The analysis of the process chain for manufacturing an electric motor is divided into a repetition part and a practical exercise part.

- Repetition of the production process chains
  - Higher-level production process chain of an electric motor
  - Production of the soft magnetic core
  - Insulation of the slot base
  - Winding process
  - Electrical contacting
  - Insulation systems
  - Permanently excited rotors
  - Mechanical components of electric motors
- **Process chain analysis based on practical examples**



Picture: VW

## Set up the process chains for the following components and process the following points

- Identify the components
- Assign the component to a function module in the electric motor
- What is the function of the component in the electrical machine?
- In which performance classes is the component used?
- Set up the process chain up to the completion of the component

# Analysis of the manufacturing process chain for electric drive components

## Example 1



# Analysis of the manufacturing process chain for electric drive components

## Example 1

### Identify the components

- Closed, insulated molded coil

### Assign the component to a function module

- Use in the stator

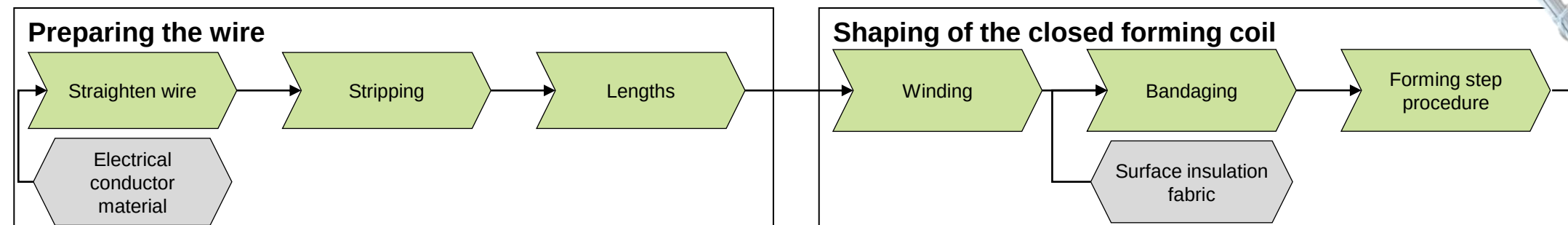
### What is the function of the component in the electrical machine?

- Conducting the electric current, generating the stator magnetic field

### In which performance classes is the component used?

- Use in high performance classes (e.g. railroad drives)

### Set up the process chain up to the completion of the component



## Analysis of the manufacturing process chain for electric drive components

### Example 2



# Analysis of the manufacturing process chain for electric drive components

## Example 2

### Identify the components

- Open form coil

### Assign the component to a function module

- Use in the stator

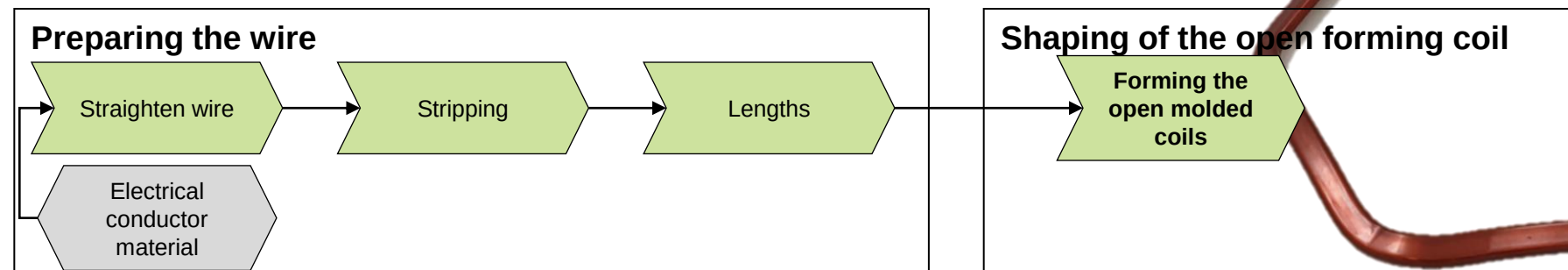
### What is the function of the component in the electrical machine?

- Conducting the electric current, generating the stator magnetic field

### In which performance classes is the component used?

- Use in medium power classes (e.g. traction drive in motor vehicles)

### Set up the process chain up to the completion of the component



## Analysis of the manufacturing process chain for electric drive components

### Example 3



# Analysis of the manufacturing process chain for electric drive components

## Example 3

### Identify the components

- Stator single tooth

### Assign the component to a function module

- Part of the stator

### What is the function of the component in the electrical machine?

- Conducting the electric current, generating the stator magnetic field

### In which performance classes is the component used?

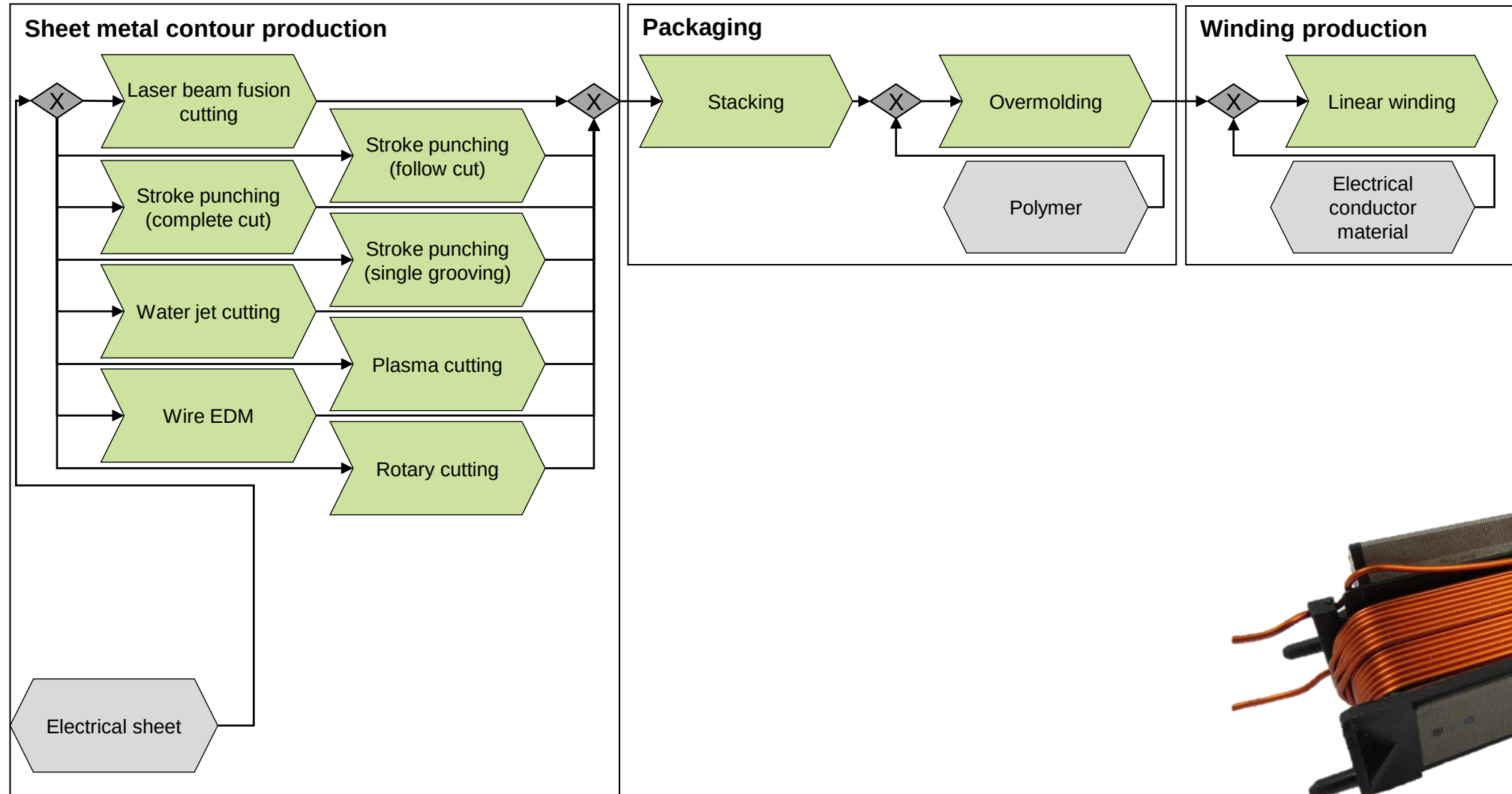
- Use in low to medium power classes (e.g. on crankshafts in HEVs)

### Set up the process chain up to the completion of the component



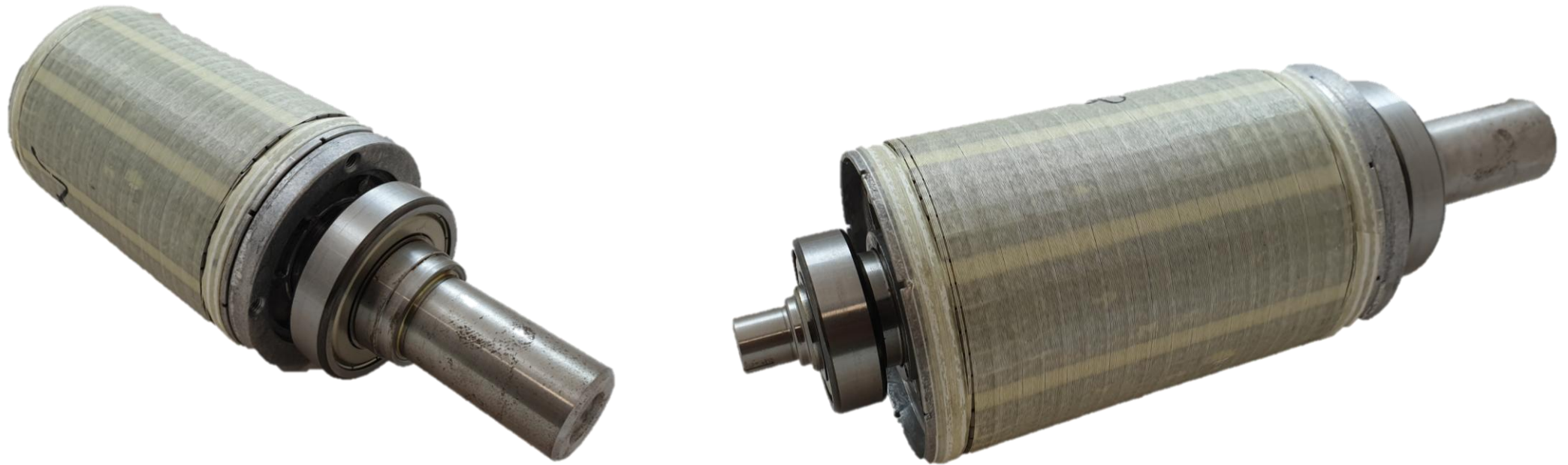
# Analysis of the manufacturing process chain for electric drive components

## Example 3



## Analysis of the manufacturing process chain for electric drive components

### Example 4



# Analysis of the manufacturing process chain for electric drive components

## Example 4

### Identify the components

- Permanently excited synchronous rotor with external permanent magnets

### Assign the component to a function module

- Rotor

### What is the function of the component in the electrical machine?

- Generating and transmitting power and torque

### In which performance classes is the component used?

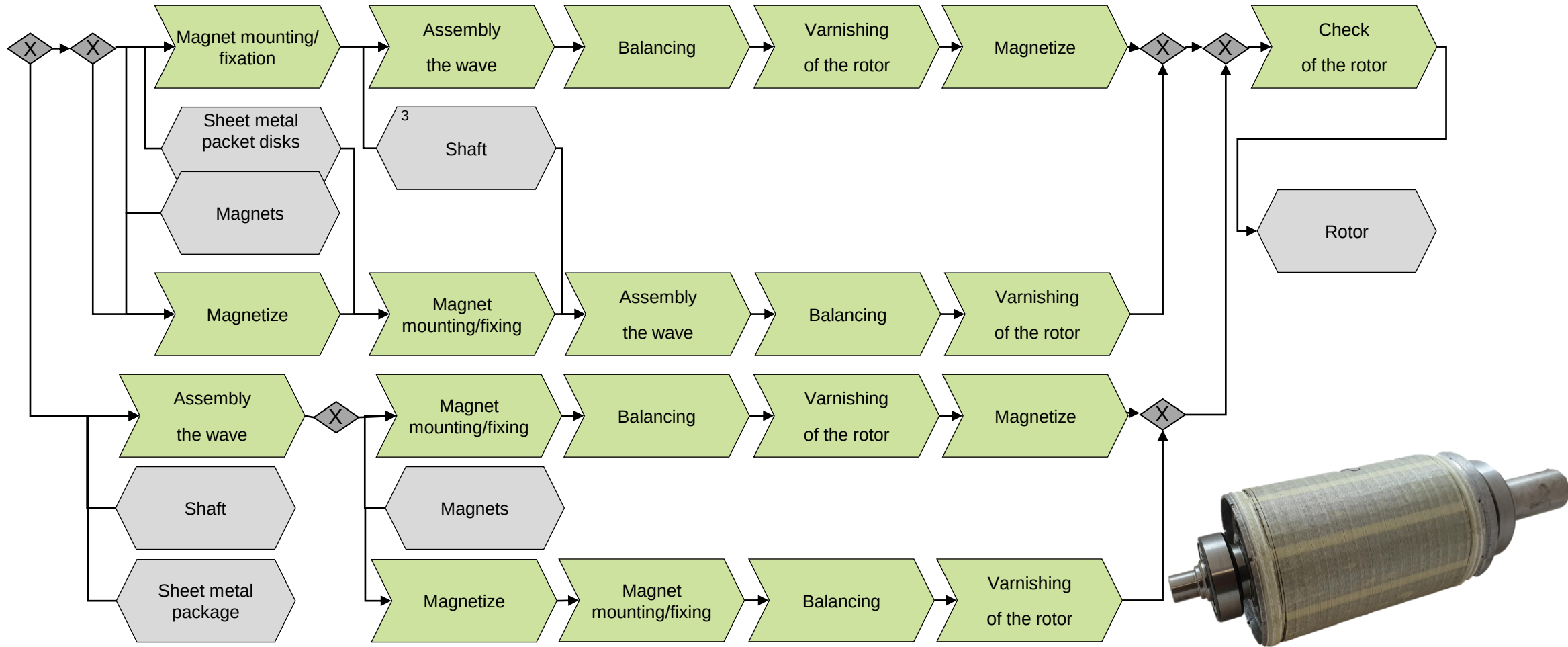
- Servo drives, applications with speeds below 10,000 rpm

### Set up the process chain up to the completion of the component



# Analysis of the manufacturing process chain for electric drive components

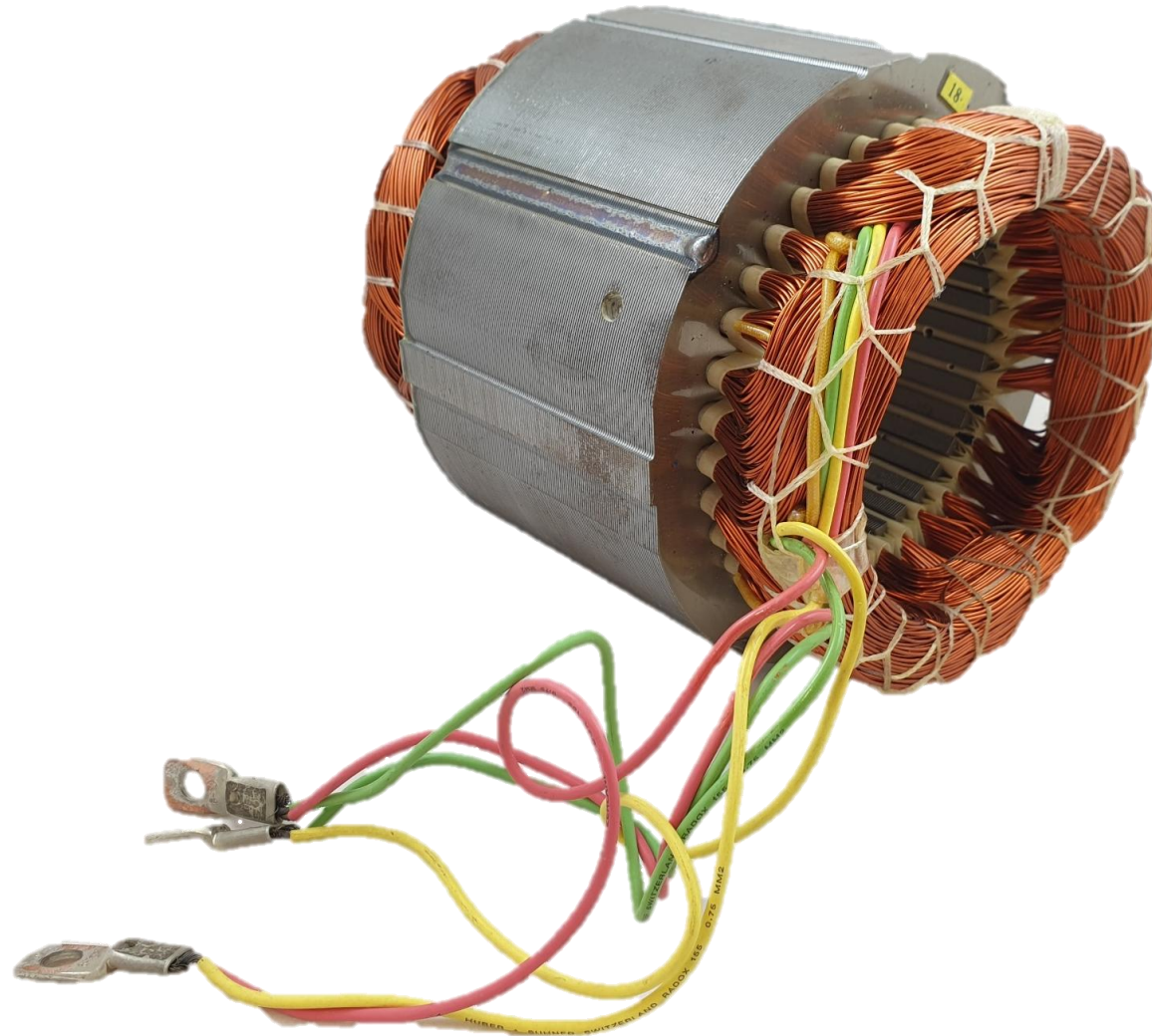
## Example 4





## Analysis of the manufacturing process chain for electric drive components

### Example 5



# Analysis of the manufacturing process chain for electric drive components

## Example 5

### Identify the components

- Conventional stator with pull-in winding

### Assign the component to a function module

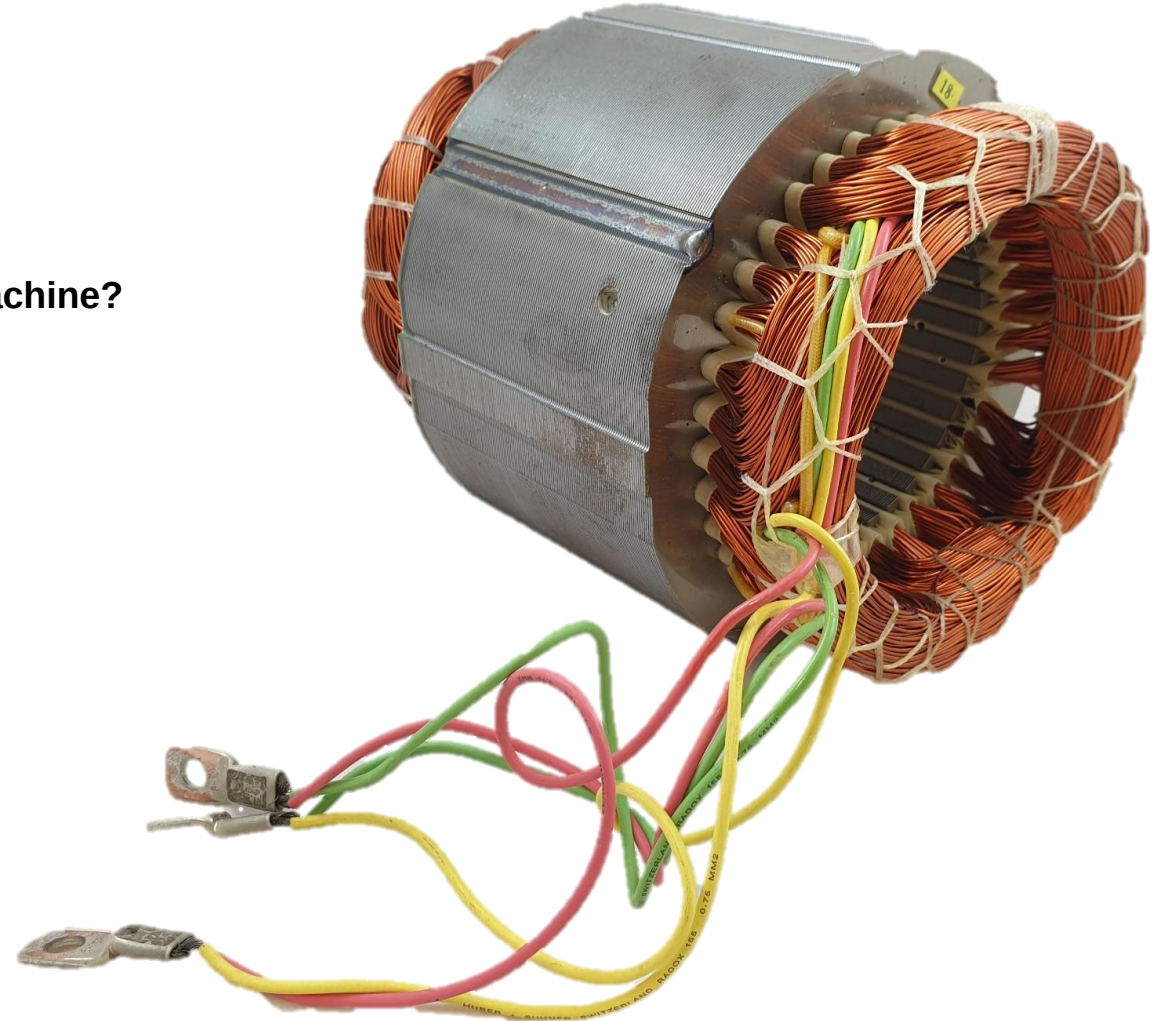
- Stator

### What is the function of the component in the electrical machine?

- Generating and transmitting the rotating magnetic field

### In which performance classes is the component used?

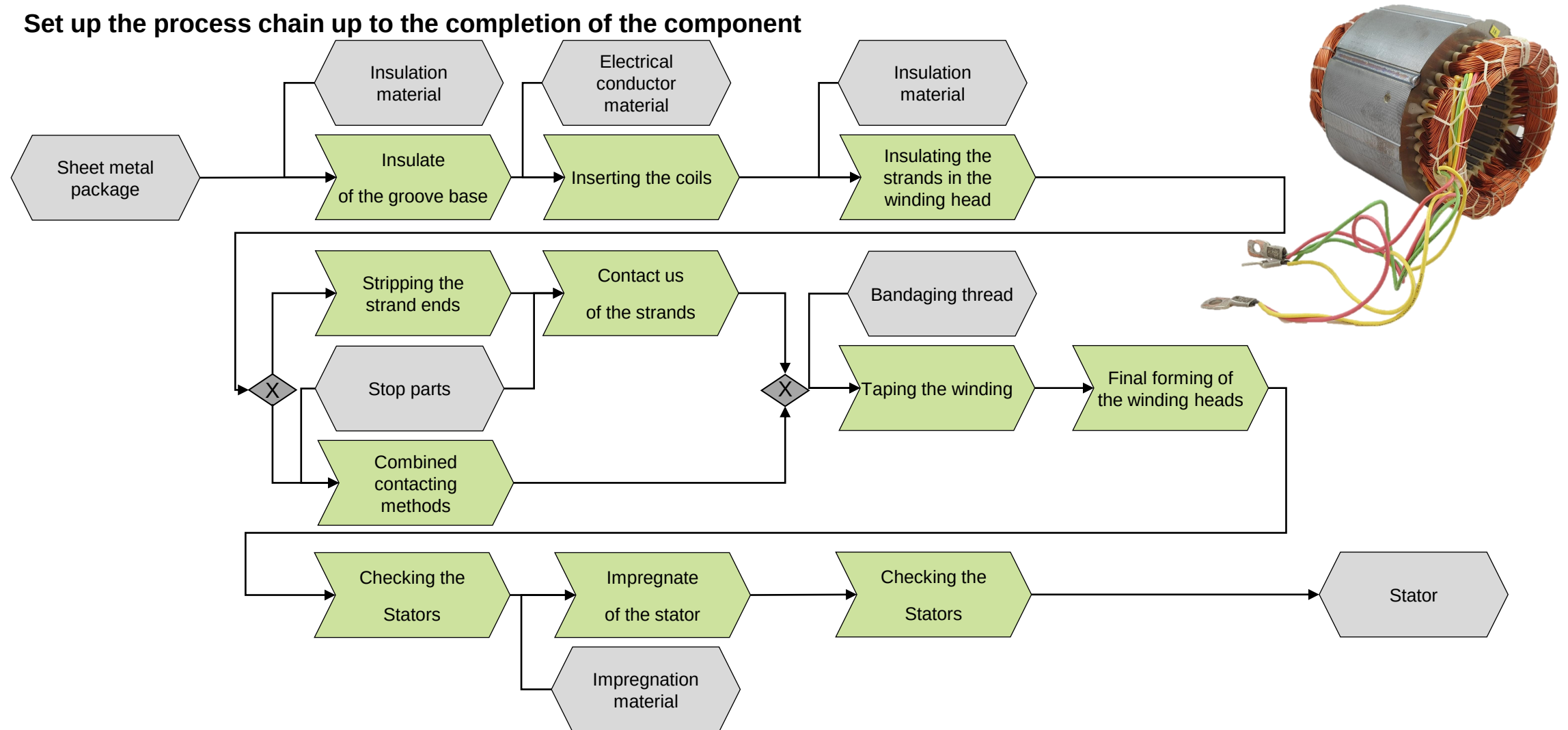
- Industrial drives, traction drives, servo drives



# Analysis of the manufacturing process chain for electric drive components

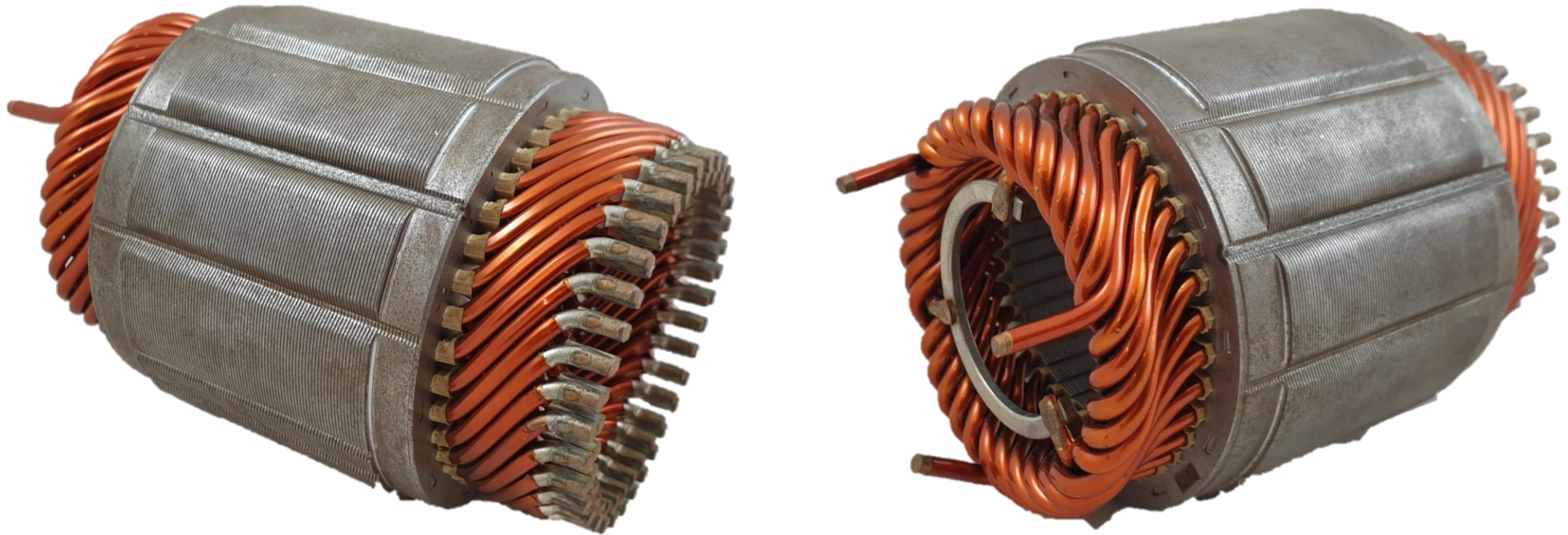
## Example 5

Set up the process chain up to the completion of the component



## Analysis of the manufacturing process chain for electric drive components

### Example 6



# Analysis of the manufacturing process chain for electric drive components

## Example 6

### Identify the components

- Stator with open form coils

### Assign the component to a function module

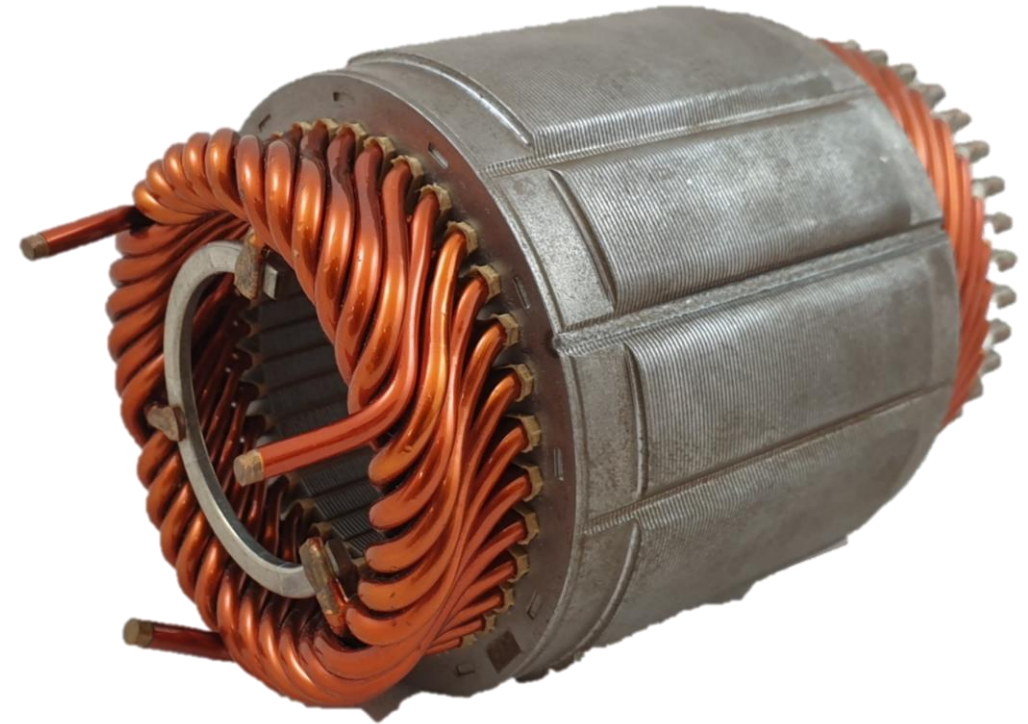
- Stator

### What is the function of the component in the electrical machine?

- Generating and transmitting the rotating magnetic field

### In which performance classes is the component used?

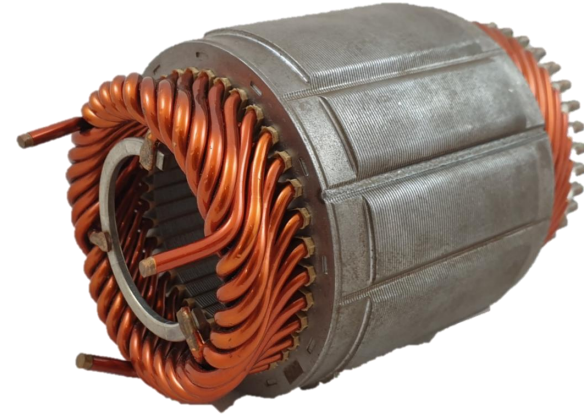
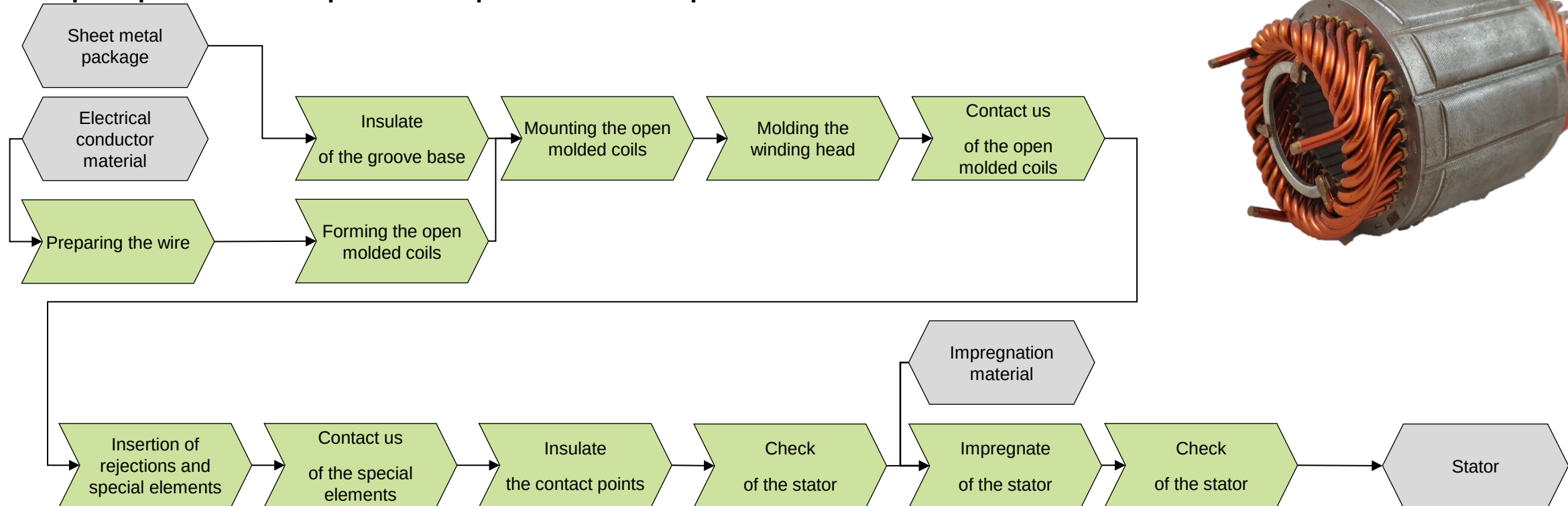
- Small to medium outputs; large quantities  
(e.g. traction drives, alternators)



# Analysis of the manufacturing process chain for electric drive components

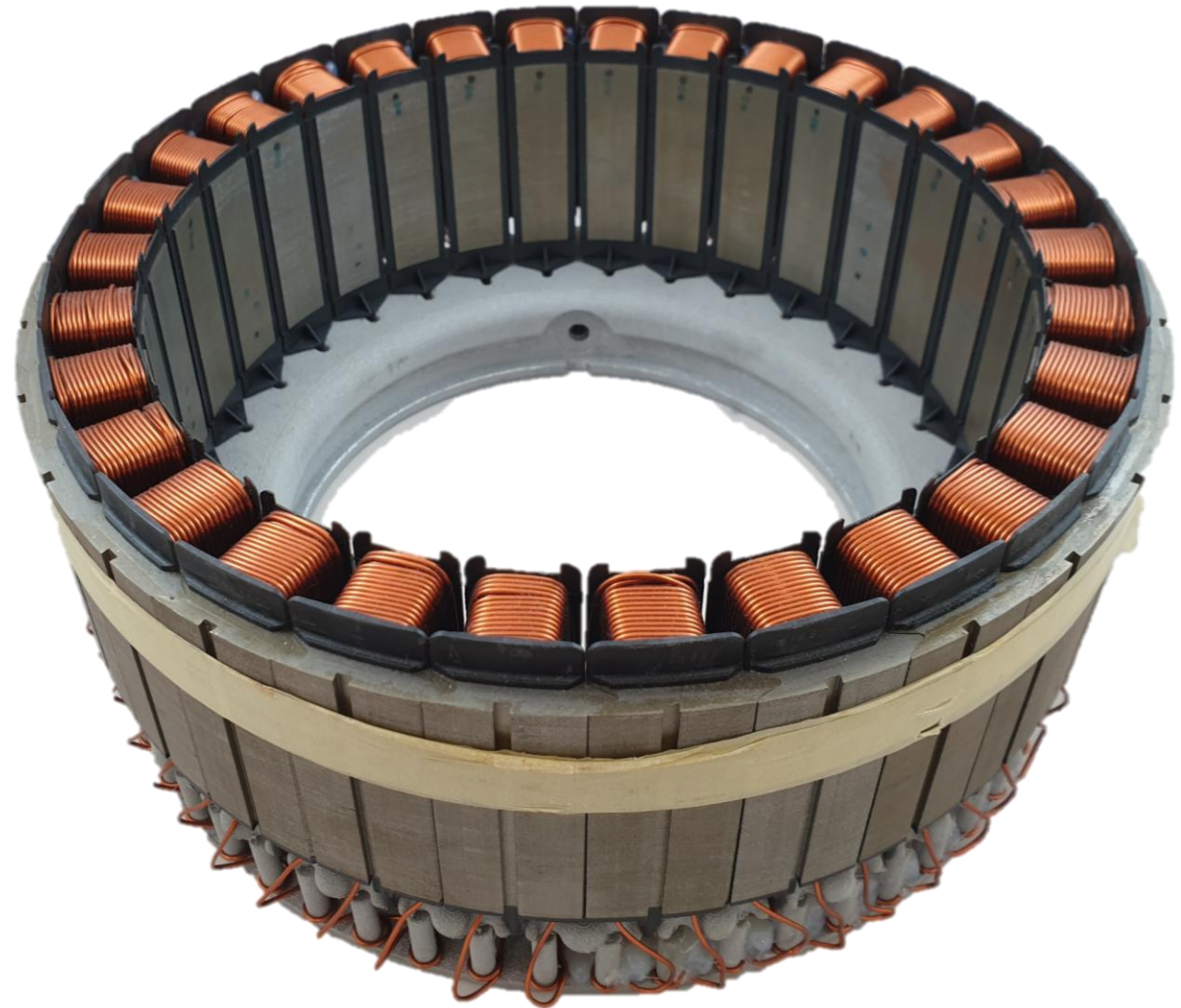
## Example 6

Set up the process chain up to the completion of the component



## Analysis of the manufacturing process chain for electric drive components

### Example 7



# Analysis of the manufacturing process chain for electric drive components

## Example 7

### Identify the components

- Segmented stator with concentrated winding

### Assign the component to a function module

- Stator

### What is the function of the component in the electrical machine?

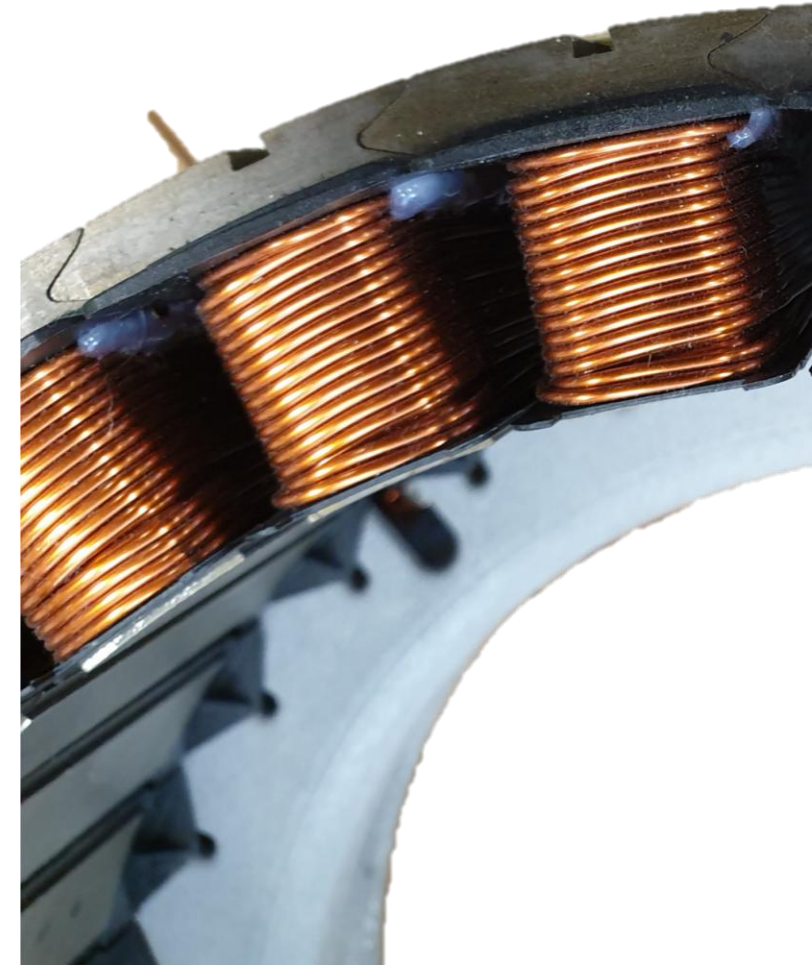
- Generating and transmitting the rotating magnetic field

### In which performance classes is the component used?

- Small to medium power; large number of units; low speed  
(e.g. mild hybrid traction drive, starter-generator)

### Set up the process chain up to the completion of the component

- See example 4



## Analysis of the manufacturing process chain for electric drive components

### Example 8



# Analysis of the manufacturing process chain for electric drive components

## Example 8

### Identify the components

- Rotor segment of a permanently excited synchronous rotor (with reluctance component) with internal / buried permanent magnets

### Assign the component to a function module

- Rotor

### What is the function of the component in the electrical machine?

- As a rotor component, interaction with stator magnetic field and participation in energy conversion

### In which performance classes is the component used?

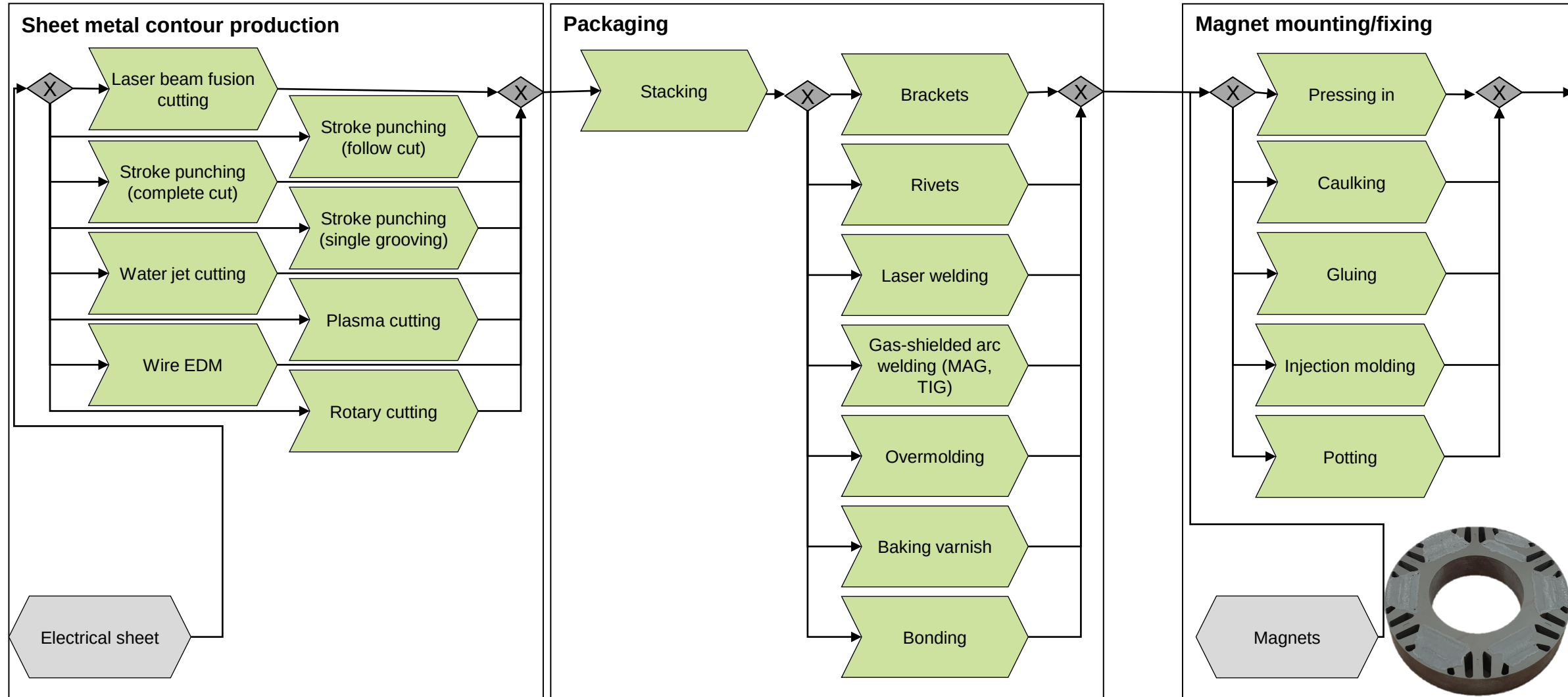
- Medium power; large quantities; high speed  
(e.g. traction drive)

### Set up the process chain up to the completion of the component



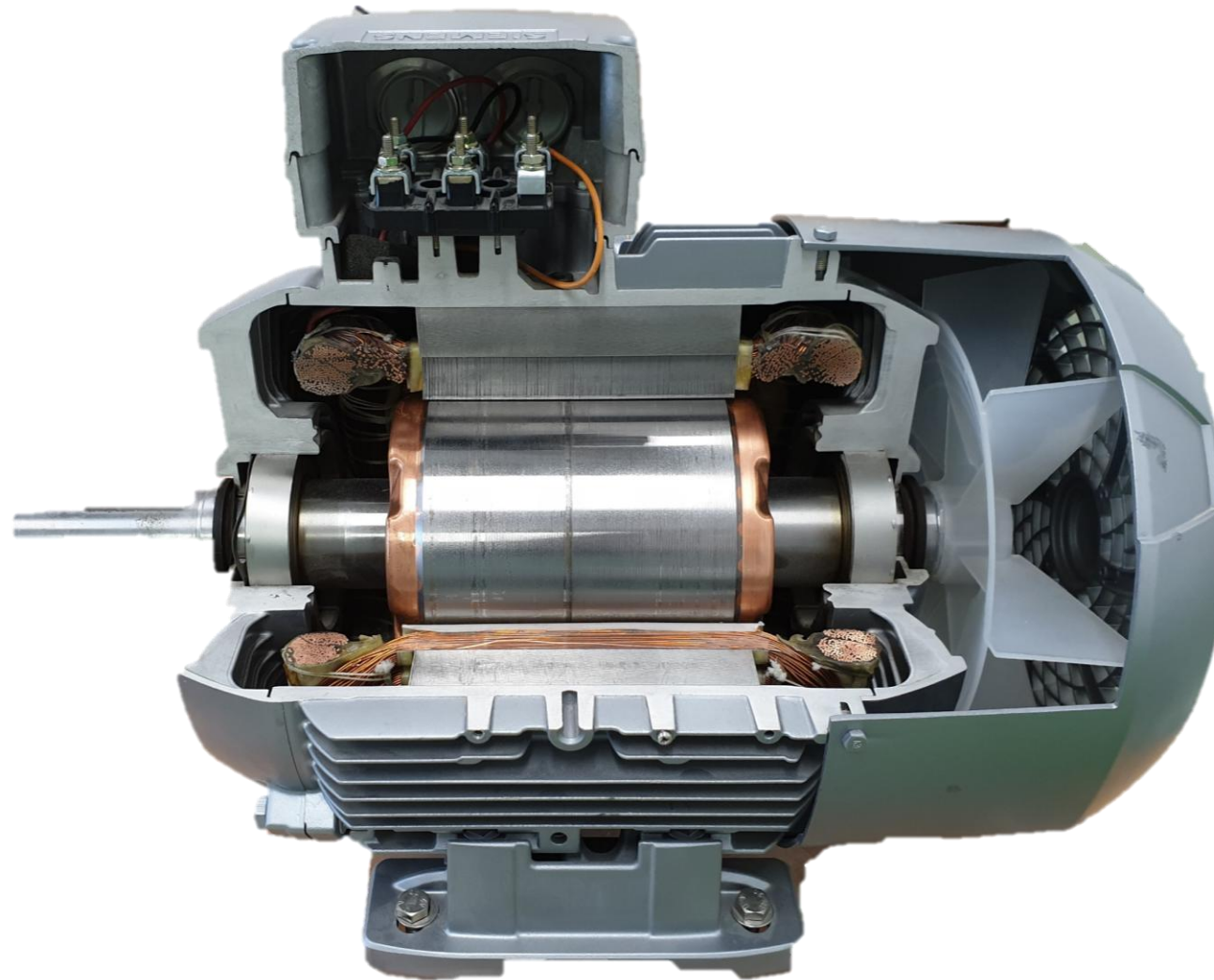
# Analysis of the manufacturing process chain for electric drive components

## Example 8



## Analysis of the manufacturing process chain for electric drive components

### Example 9



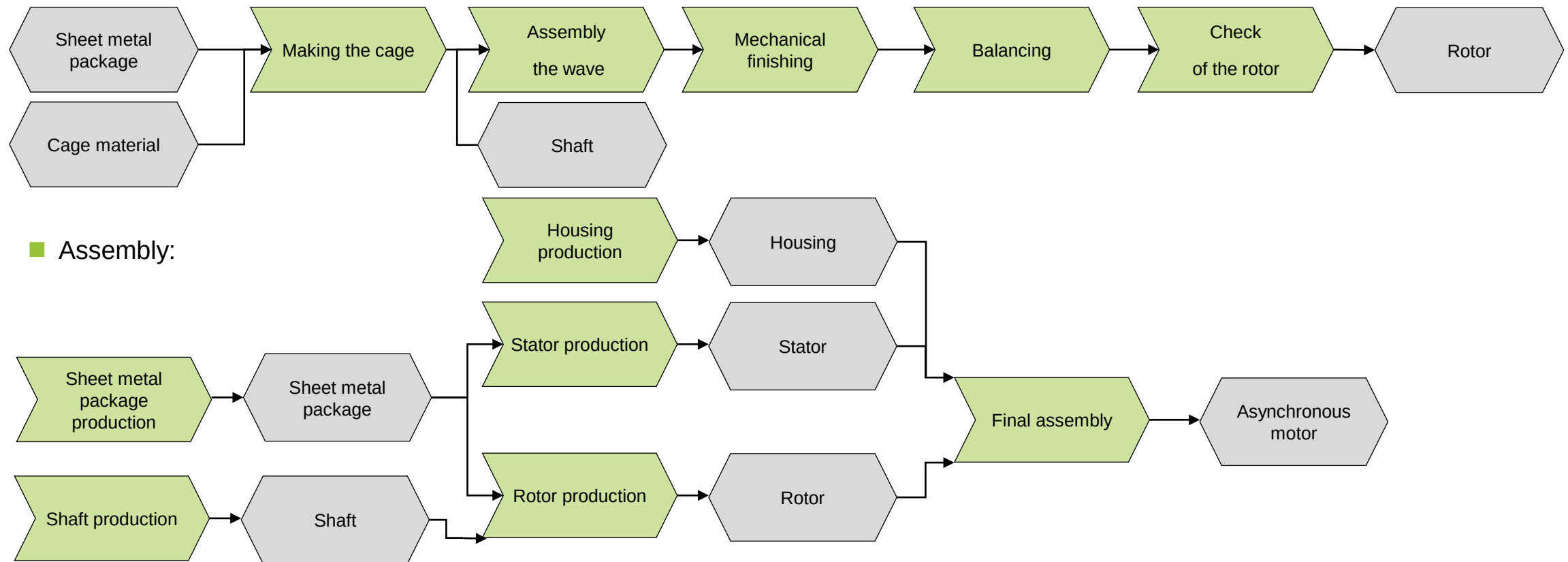
# Analysis of the manufacturing process chain for electric drive components

## Example 9

Set up the process chain up to the completion of the component

■ Stator: see Conventional stator with pull-in winding

■ Rotor:



The contents of the lecture unit can be further deepened with the following specialist literature:

### Reference books

- Much W.; et al (1986): "Wicklungen un Montage rotieren er elektrischer Maschinen", VEB Verlag Technik, Berlin, ISBN 3-341-00020-8.
- Bălă C.; et al. (1976): "Han buch er Wickeltechnik elektrischer Maschinen", VEB Verlag Technik, Berlin.
- Kampker, A. (2014): Electromobile production. Berlin: Springer Vieweg. ISBN 978-3-642-42022-1

### Dissertations

- Gläßel T. (vorr. 2020): Process chains for laser beam welding of flat conductor-based shaped coil windings for automotive traction drives. Erlangen: FAU-Press.





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# DANKE